Elementary Particles

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Introduction

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1 . Overview to Particle Physics

1.1 Elementary particles and interactions: the energy-scale viewpoint

The basic questions of particle physics are deceptively simple:

- ◇ What are the elementary building blocks of Nature?
- ◇ What are the interactions among them?

A key lesson is that the answer depends on *the energy scale* at which we probe Nature: as we increase the typical energy (or momentum transfer) of our experiments, new degrees of freedom become relevant and the effective description changes.

A pragmatic (and historically successful) way to organize our knowledge is to relate *energy scales* to the *particles we can resolve*, the *interactions we observe*, their *ranges*, and their *mediators*. At very low energies we describe matter in terms of atoms and effective potentials; at higher energies we resolve nuclei, then nucleons, and eventually quarks, leptons and gauge bosons.

1.1.1 A qualitative "energy ladder"

Scale	Degrees of freedom	Interactions	Range
$\lesssim$ eV	atoms (effective)	effective (atomic/molecular)	long
$\sim$ keV	e^-, nuclei	e.m.	long
$\sim$ MeV	$e^\pm$, p, n, ν_e	e.m. + nuclear + weak	long / short / contact
$\sim$ GeV	$\mu^\pm$, ν_μ, hadrons	e.m. + strong + weak	long / short / contact
$\sim$ 10 GeV	$\tau^\pm$, ν_τ, u, d, s, c, b	e.m. + QCD + weak	long / short / contact
$\sim$ TeV	$W^\pm$, Z^0, t, H	e.m. + QCD + EW	long / short / short

1.2 Natural units and physical intuition

At subatomic scales we inevitably need quantum mechanics, hence $\hbar$ appears in formulas. At high velocities we need special relativity, hence c appears as well. To avoid a constant

proliferation of $\hbar$ and c, we adopt *natural units*

$$\hbar = c = 1, \tag{1.1}$$

so that any dimensionful quantity can be expressed as a power of energy. In this convention, masses, momenta and energies share the same units (eV, MeV, GeV, ...).

To recover SI units, one reinserts appropriate powers of $\hbar$ and c and uses 1 eV $\simeq 1.6 \times 10^{-19}$ J. Useful reference scales include

- $E_{\text{Rydberg}} = 13.6$ eV,
- $m_e \simeq 0.5$ MeV,
- $m_p \simeq 940$ MeV.

For distances and cross sections, it is common to use

- the fermi (fm): 1 fm $= 10^{-15}$ m, and $\hbar c \simeq 197$ MeV fm,
- the barn (b): 1 b $= 100$ fm^2.

1.3 The Standard Model and what lies beyond

The Standard Model (SM) describes elementary particles and their interactions up to energies of order $\mathcal{O}(10 \text{ TeV})$. It contains:

- **Matter fields** (spin $J = \frac{1}{2}$): quarks and leptons. Charged leptons and quarks are massive; neutrino masses are very small.
- **Gauge bosons** (spin $J = 1$): photon and gluons are massless; $W^{\pm}$ and Z^0 are massive.
- **Higgs boson** (spin $J = 0$), massive.

Despite its success, there are clear indications that the SM is not the final story:

- **Gravity:** extremely weak at subatomic distances but long-range, suggesting a massless mediator (graviton) in a quantum description.
- **Dark matter:** gravitational evidence at galactic/cosmological scales with no electromagnetic signals, suggesting at least one new particle.

1.4 Energy frontiers in Nature

The highest man-made energies are achieved in colliders. At present, the LHC reaches ~ 13 TeV in the center-of-mass energy. Future proposals include FCC(hh) at ~ 100 TeV and lepton/hadron options at intermediate energies.

Nature also provides *cosmic rays* with energies far beyond colliders: ultra-high energy cosmic rays satisfy $E > 10^4$ TeV, with record events around $E \sim 3 \times 10^8$ TeV (Utah, 1991).

1.5 QFT as the language of particle physics

Combining quantum mechanics with special relativity naturally leads to *quantum field theory* (QFT):

- states can contain an *arbitrary number of particles* (Fock space),
- elementary particles correspond to *unitary irreducible representations* of the Poincaré group $\mathrm{ISO}(3,1)$.

The irreducible unitary representations are characterized by

$$m^2 \in \mathbb{R}_+, \qquad \mathbf{p} \in \mathbb{R}^3, \qquad J \in \tfrac{1}{2}\mathbb{Z}_+, \qquad J_3 \in \{-J, -J+1, \ldots, J\}, \tag{1.2}$$

with the special feature that for massless particles only the extreme spin projections appear.

1.6 Strength of interactions and unification scales

A convenient way to compare interactions is through dimensionless couplings (or effective dimensionless combinations at a given scale). At $E \sim m_p \sim 1$ GeV (low energies) and $E \sim m_Z \sim 100$ GeV (high energies), one finds qualitatively:

- **Strong:** $\alpha_s(m_p) \sim 1$ but $\alpha_s(m_Z) \sim 0.1$ (running).
- **Electromagnetic:** $\alpha(m_p) \simeq 1/137$, $\alpha(m_Z) \simeq 1/129$.
- **Weak:** at low energies appears through $G_F m_p^2 \sim 10^{-5}$, while at m_Z one may define $\alpha_W(m_Z) = \dfrac{m_W^2 G_F}{4\pi} \sim \mathcal{O}(1)$.
- **Gravity:** $G m_p^2 \sim 10^{-39}$, negligible in particle physics.

Two important conceptual scales follow from extrapolations:

- **Electroweak unification:** around $E \sim m_Z$, electromagnetic and weak strengths become comparable.
- **Grand unification:** strong and electroweak strengths may meet around $E \sim 10^{16}$ GeV.
- **Planck scale:** gravity becomes comparable to other interactions at $E \sim G^{-1/2} \sim 10^{19}$ GeV.

1.7 Conservation laws and discrete symmetries

Conservation laws are deeply tied to symmetries:

- **Space-time symmetries:** energy–momentum conservation; angular momentum conservation.
- **Internal symmetries:** electric charge; baryon number B; lepton number L.

In the SM, individual lepton flavor numbers (electron, muon, tau) are conserved at the Lagrangian level, but neutrino oscillations show this is not an exact symmetry in Nature. Quantum effects also violate B and L separately at an unobservably small level, leaving $B+L$ as the truly conserved combination within the SM description at accessible energies.

Strong and electromagnetic interactions conserve additional discrete symmetries: parity (P), charge conjugation (C), time reversal (T), and (approximately) flavor.

1.8 Hadrons: mesons, baryons and color

Although quarks and gluons are the building blocks of the strong interaction in the SM, in Nature we observe only *hadrons*: bound states that are color singlets. The simplest hadrons are:

- **Mesons:** quark–antiquark ($q\bar{q}$),
- **Baryons:** three quarks (qqq).

1.8.1 Isospin and light mesons/baryons

Many light hadrons can be organized into representations of SU(2) isospin. For example, pions form an isotriplet (π^+, π^0, π^-) and nucleons form an isodoublet (p, n). A simple quark-model picture for light mesons treats them as $q\bar{q}$ with

$$J = \tfrac{1}{2} \otimes \tfrac{1}{2} = 0 \oplus 1, \qquad I = \tfrac{1}{2} \otimes \tfrac{1}{2} = 0 \oplus 1, \tag{1.3}$$

with u and d quarks carrying charges $Q_u = \frac{2}{3}e$ and $Q_d = -\frac{1}{3}e$.

Similarly, light baryons can be described as three-quark states with vanishing orbital angular momentum:

$$\tfrac{1}{2} \otimes \tfrac{1}{2} \otimes \tfrac{1}{2} = \tfrac{1}{2} \oplus \tfrac{1}{2} \oplus \tfrac{3}{2}, \tag{1.4}$$

leading to, for instance, the nucleon doublet and the Δ quartet.

1.8.2 The Pauli principle and the need for color

A puzzle arises because the spin-3/2 Δ states are totally symmetric under exchange of quarks, apparently conflicting with the Pauli principle. The resolution is the introduction of a new quantum number: **color**. Quarks transform in the fundamental representation $\mathbf{3}$ of $\mathrm{SU}(3)_{\mathrm{color}}$, and physical states must be color singlets.

Group-theory products show why mesons and baryons can be colorless, while isolated quarks and gluons cannot appear as asymptotic states:

$$\mathbf{3} \otimes \mathbf{3}^* = \mathbf{1} \oplus \mathbf{8}, \qquad \mathbf{3} \otimes \mathbf{3} = \mathbf{3}^* \oplus \mathbf{6}, \qquad \mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3} = \mathbf{1} \oplus \mathbf{8} \oplus \mathbf{8} \oplus \mathbf{10}. \tag{1.5}$$

Hence $q\bar{q}$ and qqq can contain singlets, but single quarks, gluons (octet), or qq states cannot be observed as free particles.

1.8.3 Exotic hadrons

QCD does not forbid hadrons beyond mesons and baryons: tetraquarks and pentaquarks are allowed. Their unambiguous identification is easier in systems with heavy quarks, where the spectrum is cleaner and thresholds are better controlled.

At low energies, the strong interaction is effectively short-range because the lightest physical excitations exchanged between hadrons are pions, with $m_\pi \sim 140$ MeV.

1.9 Weak interactions and flavor quantum numbers

Hadron resonances such as ρ, ω or Δ have typical decay widths $\Gamma \sim 1$–100 MeV, characteristic of strong decays. However, the lightest isospin multiplets $(\pi^\pm, \pi^0, p, n)$ have much smaller widths:

- π^0 decays electromagnetically with $\Gamma \sim 10^{-5}$ MeV,
- $\pi^\pm$ and n decay weakly with Γ as small as 10^{-14}–10^{-24} MeV.

Weak interactions are also needed to describe muon decay and strange-hadron decays.

1.9.1 Strangeness

Strange hadrons are produced in strong interactions as particle–antiparticle pairs but decay via the weak interaction. This motivates an approximately conserved quantum number in strong processes: **strangeness** S. Strange mesons and baryons populate isospin multiplets, and the lightest ones decay weakly.

1.10 Discrete symmetry violation and CP

Parity (P) and charge conjugation (C) are maximally violated in weak interactions. A further, smaller violation of CP is observed in neutral kaon decays. This will play a central role in flavor physics and in understanding the origin of matter–antimatter asymmetry.

1.11 More generations: charm, bottom and top

The existence of the charm quark was predicted (around 1970) because flavor-changing neutral currents were not observed. Charm was discovered in 1974 through the resonance $e^+e^- \to \gamma^* \to J/\psi$, interpreted as a $c\bar{c}$ bound state.

Bottom was discovered in 1977 through $e^+e^- \to \gamma^* \to \Upsilon$, interpreted as a $b\bar{b}$ bound state. The electroweak theory then implied a third-generation partner, the top quark, and the bottom–top structure is crucial for understanding CP violation in kaon (and later B) systems.

The top quark was discovered in 1995 at the Tevatron ($p\bar{p}$ collider), with $m_t \simeq 172$ GeV and a large width $\Gamma_t \sim 1.4$ GeV. It decays weakly before hadronizing, making it unique

among quarks.

1.12 Summary and "big picture"

Particle physics is organized around:

- the energy-scale dependence of effective degrees of freedom,
- QFT as the consistent framework combining QM and SR,
- gauge symmetries and their associated forces,
- confinement and color singlets in QCD,
- weak interactions, flavor, and discrete symmetry violation.

These themes will guide the construction of concrete field theories and the analysis of real processes in the next chapters.

2 . Fields for Free Particles and Discrete Symmetries

2.1 Overview: from relativistic wave equations to QFT

We need a formalism that:

- ◇ combines Quantum Mechanics (QM) and Special Relativity (SR),
- ◇ describes processes with an arbitrary number of particles.

The outcome is *Quantum Field Theory* (QFT). A systematic way to build a QFT for free particles is:

1. Find a suitable (relativistic) wave equation.
2. Find a Lagrangian whose Euler–Lagrange equations reproduce it.
3. Apply canonical quantization rules.

In QFT, wave functions become *operators*, and the arbitrary coefficients in general solutions become *creation and annihilation operators*. We will go through:

- ◇ Schrödinger field (non-relativistic),
- ◇ scalar field (spin 0, Klein–Gordon),
- ◇ Dirac field (spin 1/2),
- ◇ vector field (spin 1, Proca/Maxwell),

and we will discuss the implementation of discrete symmetries P, C and T in each case.

2.2 The Schrödinger field (non-relativistic)

2.2.1 Equation of motion and Lagrangian

The Schrödinger equation is

$$\left(i\frac{\partial}{\partial t} + \frac{\nabla^2}{2m}\right)\psi(t,\mathbf{x}) = 0. \tag{2.1}$$

If the particle has spin s, each of the $2s+1$ components satisfies the same equation. This equation follows from the action $S = \int dt\, L$, with

$$L = \int d^3x\, \mathcal{L}, \qquad \mathcal{L}(t,\mathbf{x}) = \psi^\dagger(t,\mathbf{x}) \left(i\partial_0 + \frac{\nabla^2}{2m} \right) \psi(t,\mathbf{x}). \tag{2.2}$$

2.2.2 Canonical quantization and Fock space

Define the canonical momenta (for components $a = 1, \ldots, 2s+1$)

$$\Pi_a(x) \equiv \frac{\partial \mathcal{L}}{\partial(\partial_0 \psi_a(x))} = i\, \psi_a^*(x), \qquad \Pi_a^*(x) \equiv \frac{\partial \mathcal{L}}{\partial(\partial_0 \psi_a^*(x))} = 0. \tag{2.3}$$

Canonical quantization at $t = 0$ promotes fields to operators and imposes

$$[\hat{\psi}_a(\mathbf{x}), \hat{\Pi}_b(\mathbf{y})] = i\, \delta^{(3)}(\mathbf{x}-\mathbf{y})\, \delta_{ab}, \qquad [\hat{\psi}_a(\mathbf{x}), \hat{\psi}_b(\mathbf{y})] = [\hat{\Pi}_a(\mathbf{x}), \hat{\Pi}_b(\mathbf{y})] = 0, \tag{2.4}$$

with the rule that for fermions commutators are replaced by anticommutators.

The general solution becomes an operator expansion

$$\hat{\psi}(\mathbf{x},t) = \int \frac{d^3p}{(2\pi)^3} \sum_{\lambda=-s}^{s} \hat{a}_\lambda(\mathbf{p})\, \chi_\lambda\, e^{-iEt+i\mathbf{p}\cdot\mathbf{x}}, \qquad E = \frac{\mathbf{p}^2}{2m}, \tag{2.5}$$

with $[\hat{a}_\lambda(\mathbf{p}), \hat{a}^\dagger_{\lambda'}(\mathbf{p}')] = (2\pi)^3 \delta^{(3)}(\mathbf{p}-\mathbf{p}')\delta_{\lambda\lambda'}$ (and commutators among $\hat{a}$ or among $\hat{a}^\dagger$ vanishing, for bosons).

Assuming a vacuum state $|0\rangle$ such that $\hat{a}_\lambda(\mathbf{p})|0\rangle = 0$, one defines one-particle states $|\mathbf{p}\,\lambda\rangle = \hat{a}^\dagger_\lambda(\mathbf{p})|0\rangle$. Multi-particle states follow similarly, spanning the *Fock space*.

2.2.3 Discrete symmetries: parity and time reversal

Parity acts as $\mathbf{x} \to -\mathbf{x}$ and $\mathbf{p} \to -\mathbf{p}$, with $\mathbf{L} \to \mathbf{L}$ and $\mathbf{S} \to \mathbf{S}$. In QFT it is implemented by a unitary operator P such that $P^2 = 1$ and

$$P\, \psi(t,\mathbf{x})\, P^{-1} = \pm\psi(t,-\mathbf{x}), \qquad P\, a_\lambda(\mathbf{p})\, P^{-1} = \pm a_\lambda(-\mathbf{p}). \tag{2.6}$$

Time reversal acts as $\mathbf{x} \to \mathbf{x}$, $\mathbf{p} \to -\mathbf{p}$, $\mathbf{L} \to -\mathbf{L}$, $\mathbf{S} \to -\mathbf{S}$. It is implemented by an *antiunitary* operator T.

2.2.4 Coupling to electromagnetism: gauge invariance and minimal coupling

The Maxwell equations in vacuum follow from

$$\mathcal{L}_{\text{EM}} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}, \qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \tag{2.7}$$

which is invariant under the gauge transformation $A_\mu \to A_\mu - \partial_\mu \theta(x)$. For a charged Schrödinger field, gauge symmetry acts as $\psi(x) \to e^{iq\theta(x)}\psi(x)$. Gauge invariance is ensured by *minimal coupling*

$$\partial_\mu \to D_\mu \equiv \partial_\mu + iqA_\mu, \tag{2.8}$$

leading schematically to $\mathcal{L} = \mathcal{L}_{\text{EM}} + \psi^\dagger \left(iD_0 + \frac{\mathbf{D}^2}{2m} + \cdots \right) \psi$. If $s \neq 0$, non-minimal couplings (e.g. magnetic moment terms) may also exist.

2.3 Scalar fields: Klein–Gordon (spin 0)

2.3.1 Wave equation and Lagrangian

The simplest relativistic wave equation is the Klein–Gordon equation

$$(\partial_\mu \partial^\mu + m^2)\, \phi(x) = 0. \tag{2.9}$$

For a complex scalar field, it follows from

$$\mathcal{L} = (\partial_\mu \phi^*)(\partial^\mu \phi) - m^2 \phi^* \phi. \tag{2.10}$$

For a real scalar field, one uses

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) - \frac{1}{2} m^2 \phi^2. \tag{2.11}$$

2.3.2 Mode expansion and particle/antiparticle operators

Upon quantization, the general solution can be written as

$$\hat{\phi}(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \left(e^{-ip\cdot x}\, \hat{a}(\mathbf{p}) + e^{ip\cdot x}\, \hat{b}^\dagger(\mathbf{p}) \right), \qquad E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2}. \tag{2.12}$$

Here $\hat{a}$ annihilates a spin-0 particle and $\hat{b}$ annihilates its antiparticle. For a real scalar field, particle and antiparticle coincide: $\hat{a} = \hat{b}$.

The vacuum $|0\rangle$ satisfies $\hat{a}(\mathbf{p})|0\rangle = \hat{b}(\mathbf{p})|0\rangle = 0$. With relativistic normalization, one often chooses factors so that $\langle \mathbf{p} \,|\, \mathbf{p}\,' \rangle = 2E_{\mathbf{p}}(2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{p}\,')$.

2.3.3 Discrete symmetries P, C and T

Parity can be implemented as

$$P\, \phi(t, \mathbf{x})\, P^{-1} = \pm\phi(t, -\mathbf{x}), \qquad P\, a(\mathbf{p})\, P^{-1} = \pm a(-\mathbf{p}), \quad P\, b(\mathbf{p})\, P^{-1} = \pm b(-\mathbf{p}), \tag{2.13}$$

so particle and antiparticle have the *same* intrinsic parity for scalars.

Charge conjugation exchanges particle and antiparticle operators:

$$C\,a(\mathbf{p})\,C^{-1} = b(\mathbf{p}), \qquad C\,b(\mathbf{p})\,C^{-1} = a(\mathbf{p}), \tag{2.14}$$

and acts on the field as $C\,\phi(x)\,C^{-1} = \phi^*(x)$. For a real scalar field, C can act as $C\phi C^{-1} = \pm\phi$ (defining C-parity).

Time reversal can be implemented as

$$T\,\phi(t,\mathbf{x})\,T^{-1} = \eta_T\,\phi(-t,\mathbf{x}), \qquad |\eta_T| = 1. \tag{2.15}$$

2.3.4 Coupling to electromagnetism

Assign the gauge transformation $\phi(x) \to e^{iq\theta(x)}\phi(x)$ and apply minimal coupling:

$$\partial_\mu \to D_\mu = \partial_\mu + iqA_\mu, \qquad \mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + (D_\mu\phi)^*(D^\mu\phi) - m^2\phi^*\phi. \tag{2.16}$$

Minimal coupling implies that particle and antiparticle carry opposite charges. For a real scalar field, only non-minimal couplings are allowed.

2.4 Dirac fields: spin 1/2

2.4.1 Dirac equation and Lagrangian

The relativistic wave equation for spin-1/2 particles is the Dirac equation

$$(i\gamma^\mu\partial_\mu - m)\,\psi(x) = 0, \tag{2.17}$$

where ψ is a complex 4-component spinor and the gamma matrices satisfy

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}. \tag{2.18}$$

The corresponding Lagrangian is

$$\mathcal{L} = \bar{\psi}\,(i\partial\!\!\!/ - m)\psi, \qquad \bar{\psi} \equiv \psi^\dagger\gamma^0, \qquad \partial\!\!\!/ \equiv \gamma^\mu\partial_\mu. \tag{2.19}$$

2.4.2 Mode expansion and quantization

The quantized field admits the standard expansion

$$\hat{\psi}(x) = \int \frac{d^3p}{(2\pi)^3}\frac{1}{\sqrt{2E_{\mathbf{p}}}}\sum_{\lambda=\pm}\left(e^{-ip\cdot x}\,u_\lambda(\mathbf{p})\,\hat{a}_\lambda(\mathbf{p}) + e^{ip\cdot x}\,v_\lambda(\mathbf{p})\,\hat{b}^\dagger_\lambda(\mathbf{p})\right), \qquad E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2}, \tag{2.20}$$

where u_λ and v_λ satisfy

$$(p\!\!\!/ - m)u_\lambda(\mathbf{p}) = 0, \qquad (p\!\!\!/ + m)v_\lambda(\mathbf{p}) = 0, \qquad p\!\!\!/ \equiv \gamma^\mu p_\mu. \tag{2.21}$$

Consistency requires *fermionic* quantization: the creation/annihilation operators satisfy anticommutation relations.

2.4.3 Gamma-matrix technology (useful identities)

A complete basis of 4×4 matrices is $\{I, \gamma^5, \gamma^\mu, \gamma^5\gamma^\mu, \sigma^{\mu\nu}\}$ with $\sigma^{\mu\nu} \equiv \frac{i}{2}[\gamma^\mu, \gamma^\nu]$. Important trace properties include:

- ◇ $\mathrm{tr}(\text{odd number of } \gamma) = 0$,
- ◇ $\mathrm{tr}(\gamma^\mu\gamma^\nu) = 4g^{\mu\nu}$,
- ◇ $\mathrm{tr}(\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma) = 4(g^{\mu\nu}g^{\rho\sigma} - g^{\mu\rho}g^{\nu\sigma} + g^{\mu\sigma}g^{\nu\rho})$,
- ◇ $\mathrm{tr}(\gamma^5\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma) = 4i\epsilon^{\mu\nu\rho\sigma}$.

2.4.4 Discrete symmetries P, C and T

A standard implementation is:

$$P\,\psi(t,\mathbf{x})\,P^{-1} = \gamma^0\,\psi(t,-\mathbf{x}), \tag{2.22}$$
$$C\,\psi(x)\,C^{-1} = i\gamma^2\,\psi^*(x) \equiv \psi^c(x), \tag{2.23}$$
$$T\,\psi(t,\mathbf{x})\,T^{-1} = \eta_T\,\gamma^1\gamma^3\,\psi(-t,\mathbf{x}), \tag{2.24}$$

with η_T a phase. One consequence is that particle and antiparticle have *opposite* parity for Dirac fields (visible already at the level of how a_λ and b_λ transform).

2.4.5 Coupling to electromagnetism

Gauge invariance under $\psi \to e^{iq\theta(x)}\psi$ is ensured by minimal coupling:

$$\partial_\mu \to D_\mu = \partial_\mu + iqA_\mu, \qquad \mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}(i\not{D} - m)\psi. \tag{2.25}$$

Particle and antiparticle have opposite electric charge.

2.5 Chirality, helicity and Weyl fermions

Define the chiral projectors

$$P_R = \frac{1+\gamma^5}{2}, \qquad P_L = \frac{1-\gamma^5}{2}, \qquad \psi_R \equiv P_R\psi, \qquad \psi_L \equiv P_L\psi, \tag{2.26}$$

with $P_R + P_L = 1$, $P_R^2 = P_R$, $P_L^2 = P_L$, and $P_RP_L = 0$.

In the massless limit (equivalently, at very high energies $E \gg m$), right- and left-handed components decouple:

$$\bar{\psi}(i\not{\partial} - m)\psi = \bar{\psi}_R i\not{\partial}\psi_R + \bar{\psi}_L i\not{\partial}\psi_L - m\bar{\psi}_R\psi_L - m\bar{\psi}_L\psi_R \xrightarrow[E\gg m]{} \bar{\psi}_R i\not{\partial}\psi_R + \bar{\psi}_L i\not{\partial}\psi_L. \tag{2.27}$$

In this regime, chirality aligns with helicity: right-handed fields describe particles with positive helicity (and antiparticles with negative helicity), while left-handed fields describe particles with negative helicity (and antiparticles with positive helicity). The two-component Weyl equations arise as the building blocks of the electroweak theory.

Discrete symmetries act non-trivially on chirality; in particular, parity exchanges left and right components.

2.6 Majorana masses and Majorana fermions

Besides the usual Dirac mass term, one can add a *Majorana mass* term of the schematic form

$$\delta\mathcal{L} = -\delta m\,(\bar{\psi}\,\psi^c + \bar{\psi}^c\,\psi), \tag{2.28}$$

which violates the global U(1) symmetry $\psi \to e^{i\theta}\psi$ and therefore does not conserve "particle number minus antiparticle number".

One can define a Majorana fermion by the condition

$$\psi = \psi^c = i\gamma^2\psi^*, \tag{2.29}$$

implying that particle and antiparticle operators coincide (roughly $a_\lambda = b_\lambda$), so ψ and ψ^* are no longer independent.

2.7 Vector fields: Proca and Maxwell (spin 1)

2.7.1 Equation of motion and Lagrangians

A massive spin-1 field B_μ satisfies the Proca equation

$$\partial_\mu B^{\mu\nu} + m^2 B^\nu = 0, \qquad B^{\mu\nu} = \partial^\mu B^\nu - \partial^\nu B^\mu. \tag{2.30}$$

For $m = 0$ this reduces to Maxwell's equations (in vacuum) for a gauge field.

For a *real* vector field the Lagrangian is

$$\mathcal{L} = -\frac{1}{4}B_{\mu\nu}B^{\mu\nu} + \frac{1}{2}m^2 B_\mu B^\mu. \tag{2.31}$$

For a *complex* vector field,

$$\mathcal{L} = -\frac{1}{2}B^*_{\mu\nu}B^{\mu\nu} + m^2 B^*_\mu B^\mu. \tag{2.32}$$

2.7.2 Mode expansion and polarizations

For $m \neq 0$, the quantized field can be expanded as

$$B_\mu(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \sum_{h=0,\pm} \left(e^{-ip\cdot x}\, \epsilon_\mu^{(h)}(\mathbf{p})\, a_h(\mathbf{p}) + e^{ip\cdot x}\, \epsilon_\mu^{(h)*}(\mathbf{p})\, a_h^\dagger(\mathbf{p}) \right), \tag{2.33}$$

with transversality $p^\mu \epsilon_\mu^{(h)}(\mathbf{p}) = 0$ and completeness relations that project onto the physical spin-1 subspace.

2.7.3 Discrete symmetries and coupling to electromagnetism

Parity, charge conjugation and time reversal can be implemented consistently; for complex fields, charge conjugation exchanges particle and antiparticle operators $a_h \leftrightarrow b_h$.

If B_μ is complex and charged, gauge invariance under $B_\mu \to e^{iq\theta(x)} B_\mu$ is ensured by minimal coupling:

$$\partial_\mu \to D_\mu = \partial_\mu + iqA_\mu, \qquad \tilde{B}_{\mu\nu} \equiv D_\mu B_\nu - D_\nu B_\mu, \tag{2.34}$$

leading to a gauge-invariant Lagrangian of the form

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2} \tilde{B}^*_{\mu\nu} \tilde{B}^{\mu\nu} + m^2 B^*_\mu B^\mu. \tag{2.35}$$

For a real vector field, only non-minimal couplings are allowed (unless the field itself is the gauge field).

2.8 Summary

The free-field construction illustrates the general QFT pattern:

- ◇ Start from a wave equation, derive it from a Lagrangian,
- ◇ quantize canonically to obtain creation/annihilation operators and Fock space,
- ◇ implement discrete symmetries via unitary/antiunitary operators,
- ◇ include interactions by adding local terms that respect the symmetries (e.g. gauge invariance via minimal coupling).

This framework will be the foundation for interacting theories and for computing physical observables in the SM.

3 . Continuum Symmetries in Particle Physics

3.1 Why symmetries matter

A *continuum symmetry* is a symmetry transformation depending smoothly on continuous parameters. In quantum field theory (QFT), symmetries play a dual role:

- **Dynamical constraints:** they restrict the allowed form of the Lagrangian (hence of the interactions).
- **Spectral organization:** they explain degeneracies and classify states into *multiplets* (irreducible representations).

Continuum symmetries are typically divided into:

- **Space-time symmetries:** transformations of x^μ (Galilean or Poincaré symmetry), leading to conservation of energy-momentum and angular momentum.
- **Internal symmetries:** transformations acting on fields without changing x^μ (e.g. phase rotations, isospin, flavor), leading to conserved internal charges.

3.2 Noether's theorem in field theory

Consider a local Lagrangian density $L(\phi_r(x), \partial_\mu \phi_r(x))$ and an infinitesimal continuous transformation

$$\phi_r(x) \to \phi'_r(x) = \phi_r(x) + \delta\phi_r(x), \qquad S[\phi] = \int d^4x\, L.$$

If the action is invariant (up to a total derivative),

$$\delta L = \partial_\mu J^\mu,$$

then, for fields satisfying the Euler–Lagrange equations, there exists a conserved current

$$\partial_\mu j^\mu = 0, \qquad j^\mu = \frac{\partial L}{\partial(\partial_\mu \phi_r)}\,\delta\phi_r - J^\mu,$$

and a conserved charge

$$Q = \int d^3\mathbf{x}\, j^0(\mathbf{x}), \qquad \frac{dQ}{dt} = 0.$$

For **internal symmetries** one often has $\delta L = 0$ (so $J^\mu = 0$), and the Noether current simplifies to

$$j^\mu = \frac{\partial L}{\partial(\partial_\mu \phi_r)}\, \delta\phi_r.$$

3.3 Simple internal symmetries: U(1) charges and particle number

3.3.1 Schrödinger field (non-relativistic)

A global phase symmetry $\psi \to e^{i\theta}\psi$ implies a conserved particle-number operator

$$N = \int d^3\mathbf{x}\, \psi^\dagger \psi,$$

so N counts the number of non-relativistic particles. Spin rotations $\psi \to e^{i\theta\cdot\mathbf{S}}\psi$ are an approximate symmetry at leading order in the non-relativistic expansion, yielding approximately conserved spin.

3.3.2 Complex Klein–Gordon field

For

$$L = \partial_\mu \phi^* \partial^\mu \phi - m^2 \phi^* \phi, \qquad \phi \to e^{i\theta}\phi,$$

the conserved current is

$$j^\mu = -i\left(\partial^\mu \phi^*\, \phi - \partial^\mu \phi\, \phi^*\right).$$

The associated conserved charge measures *particles minus antiparticles* (after normal ordering removes an ill-defined vacuum constant).

3.3.3 Dirac field

For

$$L = \bar{\psi}(i\gamma^\mu D_\mu - m)\psi, \qquad \psi \to e^{i\theta}\psi,$$

the Noether current is the familiar vector current

$$j^\mu = \bar{\psi}\gamma^\mu \psi,$$

and the conserved charge again counts particles minus antiparticles.

3.3.4 Chiral symmetries at high energy

In the limit $m \simeq 0$, the Dirac Lagrangian approximately splits into independent right- and left-handed parts:

$$L \simeq \bar{\psi}_R i\gamma^\mu D_\mu \psi_R + \bar{\psi}_L i\gamma^\mu D_\mu \psi_L.$$

This yields an approximate $U(1)_R \times U(1)_L$ symmetry and two corresponding conserved chiral currents, which can be interpreted in terms of helicity-dependent number differences.

3.4 From symmetries to multiplets: representations and degeneracies

Let G be a symmetry group with conserved generators Q_a. Conservation implies

$$[H, Q_a] = 0 \quad \Rightarrow \quad \hat{g}(\theta) H \hat{g}^{-1}(\theta) = H,$$

so if $H|\alpha\rangle = E|\alpha\rangle$, then all states obtained by acting with the group,

$$|\beta\rangle = \hat{g}(\theta)|\alpha\rangle,$$

have the same energy E. Hence, **symmetries imply degeneracies**, and classifying degeneracies is the same as classifying **irreducible representations** (irreps).

A standard way to build states is to apply operators $\hat{O}_\alpha$ to the vacuum:

$$|\alpha\rangle = \hat{O}_\alpha|0\rangle, \qquad \hat{g}(\theta)\hat{O}_\alpha\hat{g}^{-1}(\theta) = M(\theta)^\beta{}_\alpha \hat{O}_\beta.$$

Two qualitatively different situations can occur:

- **Unbroken symmetry:** $\hat{g}(\theta)|0\rangle = |0\rangle$ (equivalently $Q_a|0\rangle = 0$), so the states form multiplets of G.
- **Spontaneously broken symmetry:** $\hat{g}(\theta)|0\rangle \neq |0\rangle$, so multiplet structure is not realized in the spectrum in the same way.

3.5 Lie groups and Lie algebras: the local structure of symmetries

A Lie group is both a group and a smooth manifold: elements near the identity can be parameterized as $g(\theta)$ with real coordinates θ_i. For matrix Lie groups, one expands near $\theta = 0$:

$$g(\theta) = \mathbf{1} + i\, T_i\, \theta_i + \cdots,$$

and the set of generators $\{T_i\}$ spans the associated **Lie algebra** $\mathfrak{g}$.

The commutator defines the algebra structure:

$$[T_i, T_j] = i f_{kij} T_k,$$

where f_{kij} are the **structure constants**. Finite-dimensional representations of G are smooth homomorphisms into $GL(m, \mathbb{C})$, and representations of $\mathfrak{g}$ are linear maps preserving commutators:

$$M([T, T']) = [M(T), M(T')].$$

3.6 Isospin: approximate SU(2) symmetry

3.6.1 Nucleon doublet and SU(2)

Proton and neutron have very similar masses. Writing the nucleon doublet

$$N = \begin{pmatrix} \psi_p \\ \psi_n \end{pmatrix},$$

the free Lagrangian can be reorganized as

$$L = \bar{N}\Big[(i\gamma^\mu \partial_\mu - m)\, I_2 - \Delta m\, \tau_3\Big] N,$$

with $m \gg \Delta m$. Neglecting the small splitting gives an approximate $U(2) \simeq U(1) \times SU(2)$ invariance:

- ⋄ the $U(1)$ piece corresponds to baryon-number conservation,
- ⋄ the $SU(2)$ piece corresponds to **isospin** conservation.

In the quark model, the origin of isospin is $m_u \simeq m_d$, so interactions involving u and d are approximately invariant under rotations in the (u, d) flavor space.

3.7 Flavor SU(3): extending isospin to include strangeness

Since strange baryons are heavier but not dramatically so, one can treat u, d, s as an approximate triplet

$$q = \begin{pmatrix} \psi_u \\ \psi_d \\ \psi_s \end{pmatrix},$$

and write the quark mass term as an SU(3)-symmetric piece plus breaking terms proportional to the diagonal generators (notably λ_3 and λ_8). In the limit $m_u \simeq m_d \simeq m_s$, the Lagrangian becomes approximately invariant under **flavor SU(3)**.

3.8 Representation theory toolbox for SU(2) and SU(3)

3.8.1 Cartan generators and weights

For a Lie algebra, choose a maximal commuting subalgebra (Cartan subalgebra). Its simultaneous eigenvalues label states.

- For $\mathfrak{su}(2)$ the rank is 1 and one uses T_3.
- For $\mathfrak{su}(3)$ the rank is 2 and one uses (T_3, T_8).

Eigenvalues under the Cartan generators are called **weights**. The pattern of weights forms the familiar weight diagrams used to organize multiplets.

3.8.2 SU(3) generators and raising/lowering operators

A standard basis for $\mathfrak{su}(3)$ uses the Gell-Mann matrices λ_a:

$$T_a = \frac{\lambda_a}{2}, \qquad a = 1, \ldots, 8, \qquad \mathrm{tr}(T_a T_b) = \frac{\delta_{ab}}{2}.$$

The Cartan generators can be chosen as

$$H_1 = T_3, \qquad H_2 = T_8.$$

In analogy with $\mathfrak{su}(2)$, one introduces raising and lowering operators E_α associated with roots α, which move you between weights in the diagram.

3.8.3 Invariant tensors and tensor methods

A powerful and concrete way to decompose tensor products is to use invariant tensors:

- For SU(3): $\delta^i{}_j$ and ϵ_{ijk} (and its conjugate).
- For SU(2): $\delta^i{}_j$ and ϵ_{ij} (and its conjugate).

They allow you to project tensors onto irreps by symmetrization/antisymmetrization and removing traces.

3.8.4 Key SU(3) tensor product decompositions

Several decompositions are central for hadron physics:

- **Mesons:** $3 \otimes 3^* = 8 \oplus 1$.
- **Diquarks:** $3 \otimes 3 = 6 \oplus 3^*$ (symmetric $\to 6$, antisymmetric $\to 3^*$).
- **Three quarks:** $3 \otimes 3 \otimes 3 = 10 \oplus 8 \oplus 8 \oplus 1$.
- **Gluon pairs:** $8 \otimes 8 = 27 \oplus 10 \oplus 10^* \oplus 8 \oplus 8 \oplus 1$.

These statements immediately answer what kinds of color/flavor singlets (physical states) can exist.

3.9 Hadrons as SU(3) multiplets

3.9.1 Meson octet and singlet

Because $3 \otimes 3^* = 8 \oplus 1$, light $q\bar{q}$ mesons appear as an octet plus a singlet. The octet fields can be arranged into a traceless Hermitian matrix M transforming as

$$M \to gMg^\dagger, \qquad g \in SU(3),$$

and an SU(3)-invariant free Lagrangian is compactly written as a trace:

$$L = \frac{1}{2}\,\mathrm{tr}\big(\partial_\mu M\,\partial^\mu M - m^2 M^2\big)\,,$$

which reproduces Klein–Gordon dynamics for each component field.

3.9.2 Baryon octet

The spin-$\frac{1}{2}$ baryon octet can be collected into a matrix B (Dirac fields) transforming similarly:

$$B \to gBg^\dagger, \qquad \bar{B} \to g\,\bar{B}\,g^\dagger,$$

with invariant free Lagrangian

$$L = \mathrm{tr}\big(\bar{B}\,(i\partial\!\!\!/ - m)\,B\big)\,.$$

This compact formulation automatically encodes the correct field content and normalization once the trace is expanded.

3.9.3 Baryon decuplet

The spin-$\frac{3}{2}$ baryon decuplet corresponds to the totally symmetric representation and is described (in flavor space) by a completely symmetric tensor Δ_{ijk}. A simple mass term has the form

$$L_m = -m\,\Delta^\dagger_{ijk}\Delta_{ijk},$$

and identifying indices $1, 2, 3 \leftrightarrow u, d, s$ maps components to the physical Δ, Σ^*, Ξ^* and Ω^- states.

3.10 Beyond mesons and baryons: exotic hadrons

Tensor-product logic also tells you when exotic color/flavor singlets are allowed:

- **Glueballs:** since $8 \otimes 8$ contains a singlet, purely gluonic bound states are allowed in principle.
- **Tetraquarks:** $(3 \otimes 3^*) \otimes (3 \otimes 3^*) = (1 \oplus 8) \otimes (1 \oplus 8)$ contains singlets beyond the "two-meson" trivial singlet, leaving room for non-trivial tetraquark states.

- **Pentaquarks:** $(3 \otimes 3 \otimes 3) \otimes (3 \otimes 3^*)$ similarly contains singlets beyond a simple baryon+meson interpretation, allowing non-trivial pentaquark configurations.

3.11 Flavor SU(3) breaking and Gell-Mann–Okubo relations

3.11.1 Spurion viewpoint: breaking along T_8

Exact flavor SU(3) is broken because m_u, m_d, m_s are not equal. If one keeps isospin as exact ($m_u = m_d$), the dominant breaking is associated with the diagonal generator T_8. Operationally, one adds to the SU(3)-invariant Lagrangian all terms linear in T_8 consistent with the field content and remaining symmetries.

3.11.2 Baryon octet masses: the Gell-Mann–Okubo (GMO) formula

For the baryon octet, the most general linear-in-T_8 correction involves two independent trace structures,

$$\delta L = -a\,\mathrm{tr}(\bar{B}BT_8) - b\,\mathrm{tr}(\bar{B}T_8B), \qquad a, b \in \mathbb{R}.$$

Expanding the traces yields linear relations among octet masses and implies the classic GMO formula

$$2(m_\Xi + m_N) = m_\Sigma + 3m_\Lambda,$$

which is remarkably accurate empirically.

3.11.3 Baryon decuplet: equal spacing rule

For the decuplet, the leading breaking gives an *equal spacing* relation,

$$m_{\Sigma^*} - m_\Delta = m_{\Xi^*} - m_{\Sigma^*} = m_\Omega - m_{\Xi^*},$$

historically important because it allowed the prediction of the Ω^- mass before its discovery.

3.11.4 Pseudoscalar mesons: a GMO-type relation

For the pseudoscalar octet, a linear T_8 mass correction leads to

$$4m_K^2 = m_\pi^2 + 3m_\eta^2,$$

which is moderately well satisfied.

3.11.5 Vector mesons: why the simplest GMO can fail

Applying the same logic to the vector octet gives

$$4m_{K^*}^2 = m_\rho^2 + 3m_\omega^2,$$

but this relation is badly fulfilled. The key missing ingredient is that

$$3 \otimes 3^* = 8 \oplus 1$$

also predicts a **singlet** vector meson in addition to the octet. Once SU(3) breaking is included, new quadratic terms can mix the octet isospin-zero state with the singlet. Therefore, the physical ω and ϕ are *mixtures* rather than pure octet/singlet states, and naive octet-only mass relations need not work.

3.12 Octet–singlet mixing and physical ω, ϕ, η, η'

3.12.1 Mixing mechanism

Including the singlet field S_μ (vector case) allows a symmetry-breaking quadratic term of the schematic form

$$\delta L_m \propto \mathrm{tr}(V_\mu T_8)\, S^\mu,$$

which mixes the singlet with the isospin-zero octet component. Diagonalizing the quadratic form defines the physical fields:

$$\omega_\mu^8 = \phi_\mu \cos\theta_V + \omega_\mu \sin\theta_V, \qquad S_\mu = -\phi_\mu \sin\theta_V + \omega_\mu \cos\theta_V.$$

Empirically, the vector mixing angle is large (close to "ideal mixing"), making ϕ approximately $|s\bar{s}\rangle$ and ω approximately $(|u\bar{u}\rangle + |d\bar{d}\rangle)/\sqrt{2}$.

In the pseudoscalar sector, an analogous diagonalization yields

$$S = \eta' \cos\theta_P + \eta \sin\theta_P, \qquad \eta_8 = -\eta' \sin\theta_P + \eta \cos\theta_P,$$

but the mixing angle is comparatively small, consistent with pseudoscalar phenomenology.

3.12.2 Phenomenological consequence: leptonic decays

Because neutral vector mesons couple electromagnetically via quark charges, mixing affects rates such as

$$V \to \gamma^* \to e^+e^-.$$

Writing the vector mesons in SU(3) matrix language, the coupling is proportional to

$$\mathrm{tr}(QV_{\mu\nu})F^{\mu\nu}, \qquad Q = e\,\mathrm{diag}\left(\frac{2}{3}, -\frac{1}{3}, -\frac{1}{3}\right),$$

so the quark composition implied by (near-)ideal mixing leads to distinctive predictions for relative leptonic widths.

3.13 Take-home messages

- **Noether's theorem** turns continuous symmetries into conserved currents and charges.
- **Representations** explain degeneracies: states in the same irrep must share quantum numbers and energies (unless the symmetry is broken).
- **Isospin SU(2)** and **flavor SU(3)** are approximate symmetries rooted in the near-degeneracy of light-quark masses.
- **Tensor product decompositions** directly predict which hadronic multiplets can exist and whether singlets (physical states) are possible.
- **SU(3) breaking along** T_8 yields robust mass relations (Gell-Mann–Okubo, equal spacing), but **octet–singlet mixing** is essential for correctly understanding the vector and pseudoscalar meson sectors.

4. The Quark Model and Effective Theories of Hadrons

4.1 Motivation: why do we need more than the quark model?

So far we have used approximate internal symmetries (isospin and flavor SU(3)) to organize hadrons into multiplets. Symmetry, however, does not by itself provide the *dynamics*: it does not explain *why* some hadrons are much lighter than others, nor does it systematically predict low-energy interactions.

A first dynamical attempt is the (constituent) non-relativistic quark model: hadrons are bound states of quasi-particles (constituent quarks) moving slowly, with masses approximately given by sums of constituent masses plus small interaction corrections. This picture is qualitatively useful for spectroscopy, but it becomes internally inconsistent in the light-hadron sector (especially because of the pion). The goal of this chapter is:

- to see *why* the non-relativistic quark model fails for light hadrons,
- to understand the pion as a (pseudo-)Goldstone boson of chiral symmetry breaking,
- to introduce the EFT logic (chiral perturbation theory) describing low-energy hadron physics.

For broader context on the quark model and hadron spectroscopy, see standard particle-physics texts such as (Halzen and Martin, 1984; Griffiths, 2008; Thomson, 2013), and for QFT background and symmetry concepts see (Peskin and Schroeder, 1995; Schwartz, 2014; Weinberg, 1995).

4.2 The non-relativistic quark model

4.2.1 Basic assumption and hyperfine interaction

The simplest dynamical assumption is:

Quarks move slowly inside hadrons, so hadron masses are roughly sums of constituent quark masses, plus relatively small interaction corrections.

To account for splittings between spin multiplets (e.g. 0^- vs. 1^- mesons and $\frac{1}{2}^+$ vs. $\frac{3}{2}^+$

baryons), the effective interaction must depend on spin. A common schematic ingredient is a hyperfine (spin–spin) Hamiltonian of the form

$$H_{\rm hf} = a \sum_{i \neq j} \frac{\mathbf{S}_i \cdot \mathbf{S}_j}{m_i m_j}, \tag{4.1}$$

where m_i are constituent masses, $\mathbf{S}_i$ are spin operators, and a is an effective constant encoding interaction details and wavefunction overlap. This structure is widely used in introductory discussions of hadron spectroscopy (Halzen and Martin, 1984; Griffiths, 2008; Thomson, 2013).

4.2.2 Mesons: $q_1\bar{q}_2$

For mesons, a schematic mass formula is

$$M_{q_1\bar{q}_2} \;=\; m_1 + m_2 + \frac{a}{m_1 m_2}\left(\frac{S(S+1)}{2} - \frac{3}{2}\right), \qquad S = \text{meson spin}, \tag{4.2}$$

where the spin factor comes from $\langle \mathbf{S}_1 \cdot \mathbf{S}_2 \rangle$. In the light SU(2)/SU(3) limit one may take $m_1 = m_2 = m$, giving

$$J^P = 0^- : \quad M = 2m - \frac{3a}{4m^2}, \tag{4.3}$$

$$J^P = 1^- : \quad M' = 2m + \frac{a}{4m^2}. \tag{4.4}$$

If we (naively) identify these with π and ρ, then

$$M' - M \;=\; \frac{a}{m^2} \;\simeq\; m_\rho - m_\pi \;\approx\; 630 \text{ MeV}. \tag{4.5}$$

In such a fit the hyperfine splitting scale is not "small" compared to typical constituent masses (one often finds m of order a few hundred MeV). This already signals a breakdown of the underlying non-relativistic hypothesis in the light sector: the "interaction correction" is comparable to (or larger than) the would-be constituent mass scale. The core issue is not the specific numbers, but the *self-consistency* of the expansion assumed by the model (Griffiths, 2008; Thomson, 2013).

4.2.3 Baryons: $q_1 q_2 q_3$

For baryons a similar schematic expression is

$$M_{q_1 q_2 q_3} \;=\; m_1 + m_2 + m_3 + a' \sum_{i \neq j} \frac{\langle \mathbf{S}_i \cdot \mathbf{S}_j \rangle}{m_i m_j}. \tag{4.6}$$

In the simplified SU(3)-symmetric limit $m_1 = m_2 = m_3 = m$, the spin dependence can be written as

$$M_{qqq} \;=\; 3m + \frac{a'}{2m^2}\left(S(S+1) - \frac{9}{4}\right), \qquad S = \text{baryon spin}. \tag{4.7}$$

For $J^P = \frac{1}{2}^+$ (nucleon) and $J^P = \frac{3}{2}^+$ (Δ),

$$J^P = \frac{1}{2}^+ : \quad M = 3m - \frac{3a'}{4m^2}, \tag{4.8}$$

$$J^P = \frac{3}{2}^+ : \quad M' = 3m + \frac{3a'}{4m^2}. \tag{4.9}$$

Hence

$$M' - M = \frac{3a'}{2m^2}. \tag{4.10}$$

Numerically, the N–Δ splitting is a few hundred MeV, again not parametrically small relative to typical constituent scales. The model can be a useful organizer of spectra, but it does not provide a controlled low-energy expansion for light hadrons (Halzen and Martin, 1984; Griffiths, 2008; Thomson, 2013).

4.2.4 Conclusion: inconsistency for light hadrons

The central message is:

- ◇ The non-relativistic quark model assumes a small interaction correction to constituent masses.
- ◇ In the light sector (where the pion is exceptionally light), the fitted "hyperfine" effects are not small compared to the masses being corrected.

Therefore, for low-energy QCD we need a different viewpoint. The correct language is: *symmetries plus effective field theory*. In particular, chiral symmetry and its breaking control the pion sector and organize interactions as an expansion in momenta and quark masses (Peskin and Schroeder, 1995; Weinberg, 1995; Schwartz, 2014; Donoghue et al., 1992).

4.3 The linear sigma model and chiral symmetry breaking

4.3.1 The pion puzzle

Experimentally,

$$m_\pi \sim 140 \text{ MeV} \ll m_\rho \sim 770 \text{ MeV}. \tag{4.11}$$

This suggests that the up and down quark masses are small compared to the QCD scale. In the limit $m_u \simeq m_d \simeq 0$:

- ◇ left- and right-handed Dirac fields decouple,
- ◇ isospin SU(2) enlarges to the **chiral symmetry**

$$SU_L(2) \otimes SU_R(2). \tag{4.12}$$

We do *not* observe approximate parity doublets throughout the hadron spectrum, so chiral symmetry cannot be realized linearly on the spectrum in an unbroken way. The

standard interpretation is that it is **spontaneously broken** by the vacuum, producing (nearly) massless Goldstone bosons: the pions (Peskin and Schroeder, 1995; Weinberg, 1995; Schwartz, 2014; Donoghue et al., 1992).

4.3.2 Mesonic realization

A convenient mesonic field realizing $SU_L(2) \otimes SU_R(2)$ is a 2×2 matrix

$$M(x) = \sigma(x) I_2 + i\,\tau \cdot \phi(x), \qquad \sigma, \phi \in \mathbb{R}, \tag{4.13}$$

where ϕ are pseudoscalars (identified with pions) and σ is a scalar. Consider the Lagrangian

$$\mathcal{L} = \frac{1}{4}\,\mathrm{tr}\big(\partial_\mu M^\dagger \partial^\mu M\big) - \frac{m^2}{4}\,\mathrm{tr}\big(M^\dagger M\big) - \lambda\left[\mathrm{tr}\big(M^\dagger M\big)\right]^2, \qquad \lambda > 0, \tag{4.14}$$

which is invariant under

$$M \to g_L\, M\, g_R^\dagger, \qquad g_L \in SU_L(2), \quad g_R \in SU_R(2), \tag{4.15}$$

with isospin SU(2) corresponding to $g_L = g_R$. This is the textbook linear sigma model realization of chiral symmetry (Peskin and Schroeder, 1995; Schwartz, 2014; Weinberg, 1995).

4.3.3 Vacuum structure and Goldstone bosons

Define the invariant

$$\rho \equiv \frac{1}{2}\,\mathrm{tr}(M^\dagger M), \tag{4.16}$$

so for constant fields the potential is

$$V(\rho) = \frac{m^2}{2}\rho + 4\lambda\rho^2. \tag{4.17}$$

If $m^2 > 0$, the minimum is at $\rho = 0$ and scalars/pseudoscalars are degenerate. If $m^2 < 0$, the minimum is at

$$0 = \frac{m^2}{2} + 8\lambda\rho \qquad \Rightarrow \qquad \rho_0 = -\frac{m^2}{16\lambda} > 0. \tag{4.18}$$

Isospin invariance selects a vacuum proportional to the identity:

$$M_0 = \sqrt{\rho_0}\, I_2 \equiv f_\pi\, I_2. \tag{4.19}$$

A convenient parameterization of the vacuum manifold is

$$M(x) = \big(f_\pi + S(x)\big)\, U(x), \qquad U(x) \in SU(2), \quad S(x) \in \mathbb{R}, \tag{4.20}$$

where $U(x)$ contains the Goldstone modes. If we focus on the Goldstone sector (set $S = 0$

at leading order), the kinetic term becomes

$$\mathcal{L}_{\text{Goldstone}} = \frac{f_\pi^2}{4}\,\mathrm{tr}\big(\partial_\mu U^\dagger \partial^\mu U\big)\,, \tag{4.21}$$

which is the non-linear sigma model.

Writing

$$U(x) = \exp\left(\frac{i\,\pi(x)\cdot\tau}{f_\pi}\right), \qquad \pi\cdot\tau = \begin{pmatrix} \pi^0 & \sqrt{2}\,\pi^+ \\ \sqrt{2}\,\pi^- & -\pi^0 \end{pmatrix}, \qquad \pi^\pm = \frac{\pi^1 \mp i\pi^2}{\sqrt{2}}, \tag{4.22}$$

and expanding shows that the π^a are massless when chiral symmetry is exact. This is the Goldstone pattern

$$SU_L(2)\otimes SU_R(2) \;\longrightarrow\; SU(2) \qquad \Rightarrow \qquad 3 \text{ Goldstone bosons.} \tag{4.23}$$

For the general QFT mechanism of spontaneous symmetry breaking and Goldstone bosons, see (Peskin and Schroeder, 1995; Schwartz, 2014; Weinberg, 1995).

4.3.4 The scalar mode and explicit breaking

In (4.20), the real field $S(x)$ is the heavy scalar mode. From the potential (for $m^2 < 0$) one finds a nonzero scalar mass scale, schematically

$$m_S^2 \sim -2m^2 > 0, \tag{4.24}$$

so S can be integrated out at sufficiently low energies.

Small quark masses $m_u \simeq m_d \equiv m_q$ explicitly break chiral symmetry while preserving isospin. At the level of the effective mesonic theory, one can represent explicit breaking through a term

$$\delta\mathcal{L} = \frac{f_\pi^2 B}{2}\,\mathrm{tr}\big(M_q\,U^\dagger + U\,M_q^\dagger\big)\,, \qquad M_q = m_q I_2, \tag{4.25}$$

where B is a low-energy constant. Expanding $U = \exp(i\pi^a\tau^a/f_\pi)$ to quadratic order gives the parametric scaling

$$m_\pi^2 \;\propto\; m_q, \qquad \Rightarrow \qquad m_\pi \;\propto\; \sqrt{m_q}, \tag{4.26}$$

so pions are light because they are pseudo-Goldstone bosons. This is one of the key low-energy lessons encoded systematically in chiral EFT (Donoghue et al., 1992; Schwartz, 2014; Thomson, 2013).

4.4 Including nucleons: why m_N does not vanish as $m_q \to 0$

Let $\psi = (p, n)^T$ be the nucleon isodoublet, with chiral projections $\psi_{L,R}$. A direct nucleon mass term $\bar{\psi}\psi$ violates chiral symmetry. However, a chiral-invariant Yukawa structure is

$$\mathcal{L}_N = \bar{\psi}_L i\partial\!\!\!/ \psi_L + \bar{\psi}_R i\partial\!\!\!/ \psi_R - d\,\bar{\psi}_L M \psi_R - d\,\bar{\psi}_R M^\dagger \psi_L. \tag{4.27}$$

When M condenses, $M \to f_\pi I_2$, this generates

$$m_N = d\, f_\pi, \tag{4.28}$$

which remains finite in the chiral limit. Therefore,

$$m_\pi \to 0 \;\; (m_q \to 0), \qquad \text{but} \qquad m_N \sim f_\pi \neq 0. \tag{4.29}$$

Physically, most of the nucleon mass is tied to the scale of spontaneous chiral symmetry breaking, not directly to the small explicit quark masses (Peskin and Schroeder, 1995; Schwartz, 2014; Donoghue et al., 1992).

At low energies, explicit breaking induces corrections analytic in m_q, so one expects at leading order

$$m_N = d f_\pi + \mathcal{O}(m_q), \tag{4.30}$$

while the pion mass begins already at order m_q as in (4.26).

4.5 The non-linear sigma model and chiral perturbation theory (EFT)

4.5.1 Low-energy regime and building blocks

At momenta well below the hadronic scale, and in particular for processes with

$$p \ll m_N, \tag{4.31}$$

pions can be produced, but most other hadrons cannot. This motivates an EFT built only out of pion degrees of freedom: **chiral perturbation theory**.

The building blocks are:

- the pion field matrix $U(x) \in SU(2)$, with $U^\dagger U = 1$,
- an explicit-breaking spurion M_q that transforms as $M_q \to g_L M_q g_R^\dagger$ and is set at the end to $M_q = m_q I_2$.

Power counting is

$$\partial_\mu \sim p \sim m_\pi, \qquad M_q \sim m_q \sim m_\pi^2 \sim p^2. \tag{4.32}$$

Then one writes *all* operators consistent with Lorentz symmetry, chiral symmetry (with spurions), parity, and charge conjugation, organized by order in p. The UV physics is

encoded in low-energy constants multiplying these operators. This is the general EFT philosophy in a concrete hadronic setting (Donoghue et al., 1992; Schwartz, 2014; Weinberg, 1995).

4.5.2 Leading order: only two universal terms

At leading order, $\mathcal{O}(p^2)$, the pion EFT has the universal form

$$\mathcal{L}_\chi^{(2)} = \frac{f_\pi^2}{4}\,\mathrm{tr}\big(\partial_\mu U^\dagger \partial^\mu U\big) + \frac{f_\pi^2 B}{2}\,\mathrm{tr}\big(M_q\,U^\dagger + U\,M_q^\dagger\big)\,. \tag{4.33}$$

Thus, at very low energies pion physics depends essentially on f_π and on one additional constant B (which sets m_π once M_q is fixed). At next-to-leading order, $\mathcal{O}(p^4)$, many additional operators appear, each with its own low-energy constant, encoding short-distance QCD effects (Donoghue et al., 1992; Schwartz, 2014).

4.6 Pion–nucleon EFT and the axial coupling g_A

4.6.1 Chiral-covariant building blocks

To include nucleons while keeping a controlled expansion for $p \ll m_N$, it is convenient to introduce $u(x)$ such that

$$U(x) = u(x)^2. \tag{4.34}$$

Define

$$v_\mu \equiv \frac{1}{2}\left(u\partial_\mu u^\dagger + u^\dagger \partial_\mu u\right), \qquad a_\mu \equiv \frac{1}{2}\left(u\partial_\mu u^\dagger - u^\dagger \partial_\mu u\right), \tag{4.35}$$

which transform as chiral-covariant vector/axial combinations. Then the leading pion–nucleon Lagrangian can be written schematically as

$$\mathcal{L}_{\pi N}^{(1)} = \bar{N}\left(i\partial\!\!\!/ + v\!\!\!/\right)N + g_A\,\bar{N}\,a\!\!\!/\gamma^5 N - m_N\,\bar{N}N + \cdots\,, \tag{4.36}$$

where the ellipsis denotes higher-order terms in the derivative/mass expansion. The coefficient multiplying the axial structure is the **axial coupling** g_A. It is not fixed by symmetry alone; it must be taken from experiment. This is a typical EFT feature: symmetry fixes the operator basis and power counting, but not the numerical values of the low-energy constants (Donoghue et al., 1992; Schwartz, 2014; Thomson, 2013).

4.6.2 Very low-energy scattering

Expanding u in pion fields yields, schematically,

$$a_\mu \simeq \frac{i}{2f_\pi}\,\partial_\mu(\pi\cdot\tau) + \cdots\,, \qquad v_\mu \simeq \mathcal{O}\left(\frac{\pi\,\partial\pi}{f_\pi^2}\right), \tag{4.37}$$

so at the lowest energies pion–nucleon interactions are governed primarily by f_π and g_A through the leading terms in (4.36). This explains why a small set of measured constants

can control a large class of low-energy processes in the pion sector (Donoghue et al., 1992; Schwartz, 2014).

4.7 Final remarks and SU(3) generalization

- ◇ The heavy scalar mode in the linear sigma model is *model-dependent*. The *universal* low-energy content is encoded in the EFT for $U(x)$ (and in the pion–nucleon EFT if baryons are included).
- ◇ The general EFT logic (separation of scales, symmetry-based operator expansion, and low-energy constants) is part of the standard QFT toolkit (Peskin and Schroeder, 1995; Schwartz, 2014; Weinberg, 1995).

Finally, the non-linear sigma model generalizes to flavor SU(3). In the idealized limit $m_u \simeq m_d \simeq m_s \simeq 0$ one has

$$SU_L(3) \otimes SU_R(3) \longrightarrow SU(3), \tag{4.38}$$

with $U(x) \in SU(3)$ written as

$$U(x) = \exp\left(\frac{i\,\lambda_a \phi^a}{f}\right), \tag{4.39}$$

so that kaons and the η appear as additional approximate Goldstone bosons. In the isospin limit, explicit breaking can be represented by a mass spurion

$$M_q = \begin{pmatrix} m_q & 0 & 0 \\ 0 & m_q & 0 \\ 0 & 0 & m_s \end{pmatrix}, \tag{4.40}$$

which generates the observed mass pattern among the pseudo-Goldstone multiplet (Thomson, 2013; Donoghue et al., 1992).

5 . QED for Leptons

5.1 QED for elementary spin-$\frac{1}{2}$ particles

Quantum Electrodynamics (QED) is the relativistic quantum field theory of a $U(1)$ gauge field (the photon) coupled to electrically charged matter. For a single elementary Dirac particle of charge q (and its antiparticle), the QED Lagrangian reads

$$\mathcal{L}_{\text{QED}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}\left(i\gamma^\mu D_\mu - m\right)\psi, \qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \qquad D_\mu = \partial_\mu + iqA_\mu. \tag{5.1}$$

At a given energy scale, the notion of "elementary" depends on what degrees of freedom can be resolved. In this chapter we focus on leptons as elementary particles.

If we have n elementary charged leptons ψ_j with charges q_j and masses m_j, then

$$\mathcal{L}_{\text{QED}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \sum_{j=1}^{n}\bar{\psi}_j\left(i\gamma^\mu D_\mu^{(j)} - m_j\right)\psi_j, \qquad D_\mu^{(j)} = \partial_\mu + iq_jA_\mu. \tag{5.2}$$

5.1.1 Global symmetries and flavor conservation

The Lagrangian is invariant under *independent global* phase rotations for each flavor,

$$\psi_j \to e^{i\theta_j}\psi_j, \qquad \theta_j = \text{constant}, \tag{5.3}$$

so there are n conserved charges. Physically: **each lepton flavor number is conserved in QED** (e.g. electron number, muon number, etc.).

5.1.2 Interaction Lagrangian and electromagnetic current

Expanding the covariant derivatives shows that the interaction is

$$\mathcal{L}_I = \sum_{j=1}^{n}\left(-q_j\bar{\psi}_j\gamma^\mu A_\mu\psi_j\right) \equiv -\sum_{j=1}^{n} j_j^\mu A_\mu \equiv -j^\mu A_\mu, \qquad j_j^\mu = q_j\,\bar{\psi}_j\gamma^\mu\psi_j. \tag{5.4}$$

Most elementary QED processes start at second order in $\mathcal{L}_I$ (one virtual photon exchange).

5.2 Lepton–lepton scattering: $e^-\mu^- \to e^-\mu^-$

Two historically important processes (for the development of QCD) are

- ⋄ elastic scattering: $e^-\mu^- \to e^-\mu^-$,
- ⋄ pair creation: $e^+e^- \to \mu^+\mu^-$,

which are related by *crossing*. We begin with $e^-\mu^- \to e^-\mu^-$.

5.2.1 Amplitude at second order in $\mathcal{L}_I$

At leading order (one-photon exchange),

$$i\mathcal{M} = \frac{i^2}{2!}\int d^4x\, \langle f|T\{\mathcal{L}_I(0)\mathcal{L}_I(x)\}|i\rangle. \tag{5.5}$$

Writing $\mathcal{L}_I = -j^\mu A_\mu$ and separating electron and muon currents,

$$i\mathcal{M} = \frac{i^2}{2}\int d^4x\, \langle f|T\{j_e^\mu(0)A_\mu(0)\; j_\mu^\nu(x)A_\nu(x)\}|i\rangle. \tag{5.6}$$

Using Wick contractions for the photon field gives the photon propagator,

$$\langle 0|T\{A_\mu(0)A_\nu(x)\}|0\rangle = \int \frac{d^4k}{(2\pi)^4}\,\frac{-ig_{\mu\nu}}{k^2+i\eta}\, e^{-ik\cdot(0-x)}, \tag{5.7}$$

so the amplitude factorizes into a propagator times current matrix elements.

5.2.2 External states and tree-level result

Let the initial and final lepton states be

$$|i\rangle = |e^-(p_A,\lambda_A)\rangle\; |\mu^-(p_B,\lambda_B)\rangle, \qquad |f\rangle = |e^-(p_1,\lambda_1)\rangle\; |\mu^-(p_2,\lambda_2)\rangle. \tag{5.8}$$

The electron and muon currents between one-particle states are

$$\langle e^-(p_1)|\, j_e^\mu(0)\, |e^-(p_A)\rangle = q_e\, \bar{u}(p_1)\gamma^\mu u(p_A), \qquad \langle \mu^-(p_2)|\, j_\mu^\nu(x)\, |\mu^-(p_B)\rangle = q_\mu\, \bar{u}(p_2)\gamma^\nu u(p_B)\, e^{-ix\cdot(p_B-p_2)}. \tag{5.9}$$

Performing the x-integral fixes the photon momentum $q \equiv p_B - p_2 = p_1 - p_A$ and yields

$$i\mathcal{M} = \frac{-ig_{\mu\nu}}{q^2}\; \bar{u}(p_1)\,(-iq_e\gamma^\mu)\,u(p_A)\; \bar{u}(p_2)\,(-iq_\mu\gamma^\nu)\,u(p_B). \tag{5.10}$$

For e^- and μ^-, $q_e = q_\mu = -e$, hence (up to an irrelevant overall phase)

$$\mathcal{M} = e^2\,\frac{\bar{u}(p_1)\gamma^\mu u(p_A)\; \bar{u}(p_2)\gamma_\mu u(p_B)}{q^2}. \tag{5.11}$$

In Mandelstam variables, $q^2 = t$ (for elastic scattering), so

$$\mathcal{M} = e^2 \, \frac{\bar{u}(p_1)\gamma^\mu u(p_A) \; \bar{u}(p_2)\gamma_\mu u(p_B)}{t}. \tag{5.12}$$

5.2.3 Spin averages, traces, and $|\mathcal{M}|^2$

If the initial beams are unpolarized and final spins are not measured, we average/sum:

$$|\mathcal{M}|^2 \;\to\; \overline{|\mathcal{M}|^2} = \frac{1}{2}\sum_{\lambda_A=\pm}\frac{1}{2}\sum_{\lambda_B=\pm}\sum_{\lambda_1=\pm}\sum_{\lambda_2=\pm}|\mathcal{M}|^2. \tag{5.13}$$

Using completeness relations and trace technology,

$$\sum_{\lambda_A,\lambda_1} \bar{u}(p_1)\gamma^\mu u(p_A) \;\; (\bar{u}(p_1)\gamma^\nu u(p_A))^* = \mathrm{tr}\Big[\gamma^\mu(\not{p}_A + m_e)\gamma^\nu(\not{p}_1 + m_e)\Big] \equiv 2L_e^{\mu\nu}, \tag{5.14}$$

$$\sum_{\lambda_B,\lambda_2} \bar{u}(p_2)\gamma^\mu u(p_B) \;\; (\bar{u}(p_2)\gamma^\nu u(p_B))^* = \mathrm{tr}\Big[\gamma^\mu(\not{p}_B + m_\mu)\gamma^\nu(\not{p}_2 + m_\mu)\Big] \equiv 2L_\mu^{\mu\nu}, \tag{5.15}$$

with

$$L_e^{\mu\nu} = 2\,(p_A^\mu p_1^\nu + p_A^\nu p_1^\mu - g^{\mu\nu}\, p_A\cdot p_1) + 2m_e^2 g^{\mu\nu}, \qquad L_\mu^{\mu\nu} = 2\,(p_B^\mu p_2^\nu + p_B^\nu p_2^\mu - g^{\mu\nu}\, p_B\cdot p_2) + 2m_\mu^2 g^{\mu\nu}. \tag{5.16}$$

Then

$$\overline{|\mathcal{M}|^2} = \frac{e^4}{t^2}\, L_e^{\mu\nu} L_{\mu\nu}^{\mu}. \tag{5.17}$$

It is convenient to express the result in terms of Mandelstam variables

$$s \equiv (p_A + p_B)^2 = (p_1 + p_2)^2, \qquad t \equiv (p_A - p_1)^2 = (p_2 - p_B)^2, \qquad u \equiv (p_A - p_2)^2 = (p_1 - p_B)^2, \tag{5.18}$$

satisfying $s + t + u = 2m_e^2 + 2m_\mu^2$. One finds

$$L_e^{\mu\nu} L_{\mu\nu}^{\mu} = 4\left[(s - m_e^2 - m_\mu^2)^2 + \frac{t^2}{2} + st\right], \tag{5.19}$$

hence

$$\overline{|\mathcal{M}|^2} = \frac{4e^4}{t^2}\left[(s - m_e^2 - m_\mu^2)^2 + \frac{t^2}{2} + st\right] \simeq \frac{4e^4}{t^2}\left[(s - m_\mu^2)^2 + \frac{t^2}{2} + st\right] \quad (m_\mu \gg m_e). \tag{5.20}$$

5.2.4 Differential cross section in the CoM frame

For a $2 \to 2$ process in the center-of-mass frame,

$$\left(\frac{d\sigma}{d\Omega}\right)_{\mathrm{CoM}} = \frac{\overline{|\mathcal{M}|^2}}{64\pi^2 s}\,\frac{p_f}{p_i}, \tag{5.21}$$

with $p_i = |\mathbf{p}_A| = |\mathbf{p}_B|$ and $p_f = |\mathbf{p}_1| = |\mathbf{p}_2|$. For $e^-\mu^- \to e^-\mu^-$ we have $p_f = p_i$.

In the **high-energy limit** ($p_i \gg m_\mu$),

$$s \simeq 4p_i^2, \qquad t \simeq -4p_i^2 \sin^2\frac{\theta}{2}, \qquad u \simeq -4p_i^2 \cos^2\frac{\theta}{2}, \tag{5.22}$$

and the squared amplitude becomes

$$\overline{|\mathcal{M}|^2} \simeq 2e^4\,\frac{s^2+u^2}{t^2} \simeq 2e^4\,\frac{1+\cos^4(\theta/2)}{\sin^4(\theta/2)}. \tag{5.23}$$

This is independent of p_i and is strongly **forward enhanced** (diverges as $\theta \to 0$), a classic signature of long-range Coulomb exchange.

5.2.5 Mott and Rutherford limits

In the regime $m_e \ll p_i \ll m_\mu$ (electron scattering off a heavy target in CoM kinematics), one recovers Mott's formula (essentially relativistic Rutherford with spin effects),

$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{CoM}} \simeq \frac{\alpha^2}{4p_i^2}\,\frac{\cos^2(\theta/2)}{\sin^4(\theta/2)}, \qquad \alpha \equiv \frac{e^2}{4\pi}. \tag{5.24}$$

In the non-relativistic limit $p_i \ll m_e$ (so $E_A \simeq m_e$), Rutherford scattering is recovered:

$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{CoM}} \simeq \frac{\alpha^2 m_e^2}{4p_i^4}\,\frac{1}{\sin^4(\theta/2)}. \tag{5.25}$$

5.3 The LAB frame (fixed-target kinematics)

The LAB frame is natural for fixed-target experiments. Take

$$p_A = (E_A, \mathbf{p}_A), \qquad p_B = (m_B, \mathbf{0}), \qquad p_1 = (E_1, \mathbf{p}_1), \qquad p_2 = (E_2, \mathbf{p}_2), \tag{5.26}$$

with $m_B = m_\mu$ for a muon target. Define the momentum transfer

$$q \equiv p_A - p_1, \qquad \nu \equiv \frac{q\cdot p_B}{m_B} = E_A - E_1 = q^0. \tag{5.27}$$

After integrating over the final target momentum, one finds a convenient expression where a delta function enforces on-shell kinematics. In the limit $E_A \gg m_e$, the delta function fixes E_1 to

$$E_1 = \frac{E_A}{1+\frac{2E_A}{m_B}\sin^2(\theta/2)}. \tag{5.28}$$

The differential cross section can be written as

$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{LAB}} = \frac{\overline{|\mathcal{M}|^2}}{64\pi^2 m_B^2}\,\frac{1}{\left(1+\frac{2E_A}{m_B}\sin^2(\theta/2)\right)^2} = \frac{\overline{|\mathcal{M}|^2}\,E_1^2}{64\pi^2 m_B^2 E_A^2}, \tag{5.29}$$

with $q^2 = t \simeq -4E_A E_1 \sin^2(\theta/2)$. In the limit $m_e \ll E_A \ll m_\mu$ one recovers the LAB version of Mott:

$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{LAB}} \simeq \frac{\alpha^2}{4E_A^2} \frac{\cos^2(\theta/2)}{\sin^4(\theta/2)}. \tag{5.30}$$

5.4 Pair creation: $e^+e^- \to \mu^+\mu^-$

This process is related to $e^-\mu^- \to e^-\mu^-$ by **crossing**. For unpolarized beams and unmeasured final spins, the squared amplitude is obtained by the substitution

$$s \to t, \qquad t \to s, \qquad u \to u, \tag{5.31}$$

leading to

$$\overline{|\mathcal{M}|^2} = \frac{4e^4}{s^2}\left[(t - m_e^2 - m_\mu^2)^2 + \frac{s^2}{2} + st\right] \simeq \frac{4e^4}{s^2}\left[(t - m_\mu^2)^2 + \frac{s^2}{2} + st\right]. \tag{5.32}$$

5.4.1 High-energy limit in the CoM frame

For $s \gg m_\mu^2$ and in the CoM frame,

$$s \simeq 4E_A^2, \qquad t \simeq -2E_A^2(1 - \cos\theta), \qquad u \simeq -2E_A^2(1 + \cos\theta), \tag{5.33}$$

one finds the classic result

$$\overline{|\mathcal{M}|^2}_{\text{CoM}} \simeq e^4\left(1 + \cos^2\theta\right), \qquad \left(\frac{d\sigma}{d\Omega}\right)_{\text{CoM}} \simeq \frac{\alpha^2}{4s}\left(1 + \cos^2\theta\right). \tag{5.34}$$

Integrating over angles gives

$$\sigma_{\text{CoM}} = \frac{4\pi\alpha^2}{3s}. \tag{5.35}$$

The angular distribution peaks at $\theta = 0, \pi$.

5.4.2 Near threshold: β-factor

Close to threshold ($s \gtrsim 4m_\mu^2$), the outgoing muons are slow. Define

$$\beta \equiv \sqrt{1 - \frac{4m_\mu^2}{s}}, \qquad p_f = \frac{\sqrt{s}}{2}\beta. \tag{5.36}$$

The result becomes (showing the leading mass correction structure)

$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{CoM}} \simeq \frac{\alpha^2\beta}{4s}\left[(1 + \cos^2\theta) + \frac{4m_\mu^2}{s}\sin^2\theta\right], \qquad \sigma_{\text{CoM}} \simeq \frac{4\pi\alpha^2\beta}{3s}\left(1 + \frac{2m_\mu^2}{s}\right). \tag{5.37}$$

5.5 Scalar pair production: $e^+e^- \to \pi^+\pi^-$ at low energy

At energies $E_A \ll 1$ GeV one may treat pions approximately as point-like scalars. Then one uses QED for the electron plus **scalar QED** for the pion field ϕ. The relevant interactions are

$$\mathcal{L}_I = -q_e\,\bar{\psi}\gamma^\mu A_\mu\psi + iq_\pi A^\mu\,(\partial_\mu\phi^*\,\phi - \phi^*\,\partial_\mu\phi) + q_\pi^2 A_\mu A^\mu\,\phi^*\phi. \tag{5.38}$$

The last term is quadratic in q_π and does not contribute at leading order for one-photon exchange.

Proceeding as before, one finds (for $E_A \gg m_e$) an angular distribution in the CoM frame of the form

$$\overline{|\mathcal{M}|^2}_{\text{CoM}} \simeq e^4\,\beta^2\sin^2\theta, \qquad \beta \equiv \sqrt{1-\frac{4m_\pi^2}{s}}. \tag{5.39}$$

Hence,

$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{CoM}} \simeq \frac{\alpha^2\beta^3}{8s}\,\sin^2\theta, \qquad \sigma_{\text{CoM}} \simeq \frac{\pi\alpha^2\beta^3}{3s}. \tag{5.40}$$

Key physics: the angular distribution peaks at $\theta = \pi/2$ for scalar production, whereas $\mu^+\mu^-$ production peaks at $\theta = 0, \pi$. Measuring angular distributions allows one to infer the spin of the produced particles.

For $s \gg m_\pi^2$, one gets $\sigma_{\pi\pi} \simeq \pi\alpha^2/(3s)$, i.e. a factor of 1/4 relative to $\sigma_{\mu\mu} = 4\pi\alpha^2/(3s)$.

5.6 Other elementary QED processes (overview)

Besides lepton–lepton scattering and pair creation, several $2 \to 2$ reactions appear repeatedly:

5.6.1 Møller scattering: $e^-e^- \to e^-e^-$

- ◇ Two tree-level diagrams contribute (exchange of identical fermions), with a **relative minus sign**.
- ◇ The final state contains two identical particles, so the cross section must be divided by 2!.
- ◇ Related by crossing to Bhabha scattering and to $e^+e^+ \to e^+e^+$.

5.6.2 Bhabha scattering: $e^-e^+ \to e^-e^+$

- ◇ Two tree-level diagrams contribute (annihilation and scattering channels), with a **relative minus sign**.
- ◇ Related by crossing to Møller scattering.

5.6.3 Compton scattering: $e^-\gamma \to e^-\gamma$

- ◇ Two diagrams contribute (the photon attaches before/after the interaction along the electron line), with the **same sign**.
- ◇ Photon polarization vectors enter explicitly in the amplitude.
- ◇ The internal line is a **Dirac propagator**.

5.6.4 Annihilation into photons: $e^-e^+ \to \gamma\gamma$

- ◇ Two diagrams contribute, with the **same sign**.
- ◇ Photon polarizations enter explicitly.
- ◇ Related by crossing to Compton scattering and to $\gamma\gamma \to e^-e^+$.
- ◇ Important for positronium (e^-e^+ bound state) decay.

5.7 Summary

QED provides a clean laboratory where:

- ◇ global flavor symmetries imply lepton flavor conservation (in pure QED),
- ◇ tree-level amplitudes follow from one-photon exchange and factorize into currents,
- ◇ spin sums reduce to gamma-matrix traces and lead naturally to expressions in Mandelstam variables,
- ◇ angular distributions and threshold factors (β) encode the spin and mass of produced particles.

These techniques will be essential when moving to non-abelian gauge theories and to processes involving hadrons.

6 . QED for Hadrons

6.1 Motivation: QED probes internal structure

In Quantum Electrodynamics (QED), elementary charged fields couple to the photon through the conserved electromagnetic (e.m.) current. For a set of elementary Dirac fermions ψ_j with charges q_j,

$$\mathcal{L}_{\text{QED}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \sum_{j=1}^{n} \bar{\psi}_j\big(i\gamma^\mu D_\mu^{(j)} - m_j\big)\psi_j, \qquad D_\mu^{(j)} = \partial_\mu + iq_j A_\mu. \tag{6.1}$$

The interaction term is

$$\mathcal{L}_I = -\sum_{j=1}^{n} q_j\, \bar{\psi}_j \gamma^\mu A_\mu \psi_j \equiv -\sum_{j=1}^{n} j_j^\mu A_\mu \equiv -j^\mu A_\mu, \qquad \partial_\mu j^\mu = 0. \tag{6.2}$$

If the external states are elementary, amplitudes are computable perturbatively in α.

Hadrons are not elementary: they are composite bound states of quarks and gluons. At sufficiently low momentum transfer the photon sees the hadron as point-like, but as the momentum transfer increases, the photon resolves the hadron's internal structure. We typically cannot compute this structure from first principles without doing QCD. Instead, we **parameterize our ignorance** in the most general way allowed by:

- ◇ space-time translation invariance,
- ◇ Lorentz covariance,
- ◇ discrete symmetries (P, C when appropriate),
- ◇ and current conservation (gauge invariance).

The result is the notion of **form factors** (for exclusive processes) and **structure functions** (for inclusive processes).

6.2 Form factors for composite hadrons

6.2.1 General logic: what you can write down

Suppose an electromagnetic process involves a hadronic matrix element of the QCD electromagnetic current

$$j^{\mu}_{\rm QCD}(x) = \sum_{f=u,d,s,\ldots} q_f\, \bar{q}_f(x)\gamma^{\mu} q_f(x). \tag{6.3}$$

The object we do *not* know how to compute in general is a matrix element such as $\langle \text{hadrons}|j^{\mu}_{\rm QCD}(x)|\text{hadrons}\rangle$. We proceed in steps:

1. **Translation invariance** fixes the x-dependence:

$$\langle f|j^{\mu}_{\rm QCD}(x)|i\rangle = e^{-i(p_f-p_i)\cdot x}\,\langle f|j^{\mu}_{\rm QCD}(0)|i\rangle.$$

2. **Lorentz covariance** constrains the tensor/spinor structures that can appear.
3. **Discrete symmetries** remove forbidden structures (e.g. parity eliminates certain γ_5 terms in vector-current matrix elements between positive-parity states).
4. **Current conservation** ($\partial_\mu j^\mu = 0$) implies $q_\mu\langle f|j^\mu|i\rangle = 0$ and further reduces the number of independent terms.

All remaining freedom is encoded in a *finite number* of scalar functions of Lorentz invariants (e.g. q^2), called **form factors**.

6.2.2 Pion electromagnetic form factor

Consider $e^-e^+ \to \pi^+\pi^-$. In the center-of-mass frame, at energies where the pion structure starts to matter, the pion couples through the quark current $j^{\mu}_{\rm QCD}$.

The unknown hadronic ingredient is

$$\langle \pi^+(p_2)\,\pi^-(p_1)|\, j^{\nu}_{\rm QCD}(x)\,|0\rangle. \tag{6.4}$$

Translation invariance gives

$$\langle \pi^+\pi^-|\, j^{\nu}_{\rm QCD}(x)\,|0\rangle = e^{-i(p_1+p_2)\cdot x}\,\langle \pi^+\pi^-|\, j^{\nu}_{\rm QCD}(0)\,|0\rangle. \tag{6.5}$$

Lorentz covariance implies the most general decomposition (with scalar functions of $s = (p_1+p_2)^2$):

$$\langle \pi^+(p_2)\,\pi^-(p_1)|\, j^{\nu}_{\rm QCD}(0)\,|0\rangle = e\,(p_1+p_2)^{\nu}\,\widetilde{F}_{\pi}(s) \;+\; e\,(p_2-p_1)^{\nu}\,F_{\pi}(s). \tag{6.6}$$

Now impose current conservation:

$$0 = (p_1+p_2)_{\nu}\langle \pi^+\pi^-|\, j^{\nu}_{\rm QCD}(0)\,|0\rangle = e\,(p_1+p_2)^2\,2\,\widetilde{F}_{\pi}(s), \tag{6.7}$$

hence $\widetilde{F}_\pi(s) = 0$. Therefore,

$$\boxed{\langle \pi^+(p_2)\, \pi^-(p_1)|\, j^\nu_{\rm QCD}(0)\, |0\rangle = e\,(p_2 - p_1)^\nu\, F_\pi(s)} \tag{6.8}$$

where $F_\pi(s)$ is the **pion electromagnetic form factor**. At low energies (near threshold) the pion behaves approximately point-like, so

$$F_\pi(s) \simeq 1 \qquad \text{for } s \gtrsim 4m_\pi^2 \text{ and small momentum transfer.} \tag{6.9}$$

Experiment measures $F_\pi(s)$; in principle QCD can predict it.

6.2.3 Spin-$\frac{1}{2}$ hadrons: proton form factors and vertex decomposition

For elastic electron–proton scattering $e^-p \to e^-p$, the hadronic ingredient is

$$\langle p(p_2, \lambda_2)|\, j^\nu_{\rm QCD}(x)\, |p(p_B, \lambda_B)\rangle. \tag{6.10}$$

Translation invariance yields

$$\langle p(p_2)|\, j^\nu_{\rm QCD}(x)\, |p(p_B)\rangle = e^{-i(p_2 - p_B)\cdot x}\, \langle p(p_2)|\, j^\nu_{\rm QCD}(0)\, |p(p_B)\rangle, \tag{6.11}$$

so define the momentum transfer $q = p_2 - p_B$, $q^2 = t$.

Lorentz covariance allows the most general Dirac decomposition consistent with a vector current. After imposing parity invariance (eliminating γ_5 structures) and current conservation ($q_\nu \langle p|j^\nu|p\rangle = 0$), the matrix element can be written in terms of two independent scalar functions:

$$\boxed{\langle p(p_2, \lambda_2)|\, j^\nu_{\rm QCD}(0)\, |p(p_B, \lambda_B)\rangle = \bar{u}_{\lambda_2}(p_2) \left[e\, F_1(q^2)\gamma^\nu + e\, \frac{F_2(q^2)}{2m_p}\, i\sigma^{\nu\rho} q_\rho \right] u_{\lambda_B}(p_B).} \tag{6.12}$$

Here:

- ◇ $F_1(q^2)$ is the **Dirac form factor** (charge distribution),
- ◇ $F_2(q^2)$ is the **Pauli form factor** (magnetization/anomalous moment),
- ◇ $\sigma^{\nu\rho} = \frac{i}{2}[\gamma^\nu, \gamma^\rho]$.

Low-q^2 normalization and anomalous magnetic moment. If the proton were a minimally coupled point Dirac particle, one would expect $F_1(0) = 1$ and $F_2(0) = 0$. Experiment instead finds a large anomalous magnetic moment. In an effective-field-theory spirit, one can include the leading non-minimal (dimension-5) Pauli term compatible with gauge invariance, Lorentz invariance and discrete symmetries:

$$\delta\mathcal{L} = \frac{e\,\kappa_p}{2m_p}\, \bar{\psi}_p \sigma^{\mu\nu} F_{\mu\nu} \psi_p, \qquad \kappa_p = \frac{g_p}{2} - 1 \simeq 1.79, \quad g_p \simeq 5.58. \tag{6.13}$$

Then the low-q^2 behavior becomes

$$F_1(0) = 1, \qquad F_2(0) = \kappa_p. \tag{6.14}$$

Similarly for the neutron: $F_1(0) = 0$ (net charge 0) and $F_2(0) = \kappa_n \simeq -1.91$.

6.2.4 Unpolarized elastic scattering: hadronic tensor and Sachs form factors

If beams and detectors are unpolarized, we do not need the full spin-dependent matrix element, only the spin-averaged product (hadronic tensor):

$$L_p^{\mu\nu}(p_B, p_2) \equiv \frac{1}{2} \sum_{\lambda_B,\lambda_2} \langle p(p_2, \lambda_2)|j^\mu_{\rm QCD}(0)|p(p_B, \lambda_B)\rangle \, \langle p(p_2, \lambda_2)|j^\nu_{\rm QCD}(0)|p(p_B, \lambda_B)\rangle^*. \quad (6.15)$$

Lorentz and parity invariance imply that $L_p^{\mu\nu}$ can be decomposed into a small basis of tensors built from $g^{\mu\nu}$, $p^\mu = (p_B + p_2)^\mu$, and $q^\mu = (p_2 - p_B)^\mu$. Current conservation removes terms proportional to q^μ in the appropriate way, leaving

$$L_p^{\mu\nu} = 2m_p^2 \, G_1(q^2) \left(-g^{\mu\nu} + \frac{q^\mu q^\nu}{q^2} \right) + G_2(q^2) \, p^\mu p^\nu, \quad (6.16)$$

so unpolarized elastic scattering depends on **two** scalar functions, consistent with the two form factors in (6.12).

It is conventional (and physically transparent) to use the **Sachs form factors** G_E and G_M, defined (with $Q^2 \equiv -q^2 > 0$ and $\tau \equiv Q^2/(4m_p^2)$) by

$$G_E(Q^2) = F_1(Q^2) - \tau F_2(Q^2), \qquad G_M(Q^2) = F_1(Q^2) + F_2(Q^2). \quad (6.17)$$

Interpretation:

- ⋄ G_E is associated with the **electric charge distribution**,
- ⋄ G_M is associated with the **magnetization distribution**.

For a point-like minimally coupled Dirac particle one expects $G_E = G_M = 1$.

Rosenbluth-type form in the LAB frame. Early electron–proton scattering experiments were performed in fixed-target kinematics (LAB). The differential cross section can be organized so that G_E and G_M may be extracted from angular dependence (Rosenbluth separation). In the notes' conventions one often finds a form

$$\left(\frac{d\sigma}{d\Omega} \right)_{\rm LAB} = \frac{\alpha^2 E_1}{4E_A^3 \sin^4(\theta/2)} \left(G_2 \cos^2 \frac{\theta}{2} + G_1 \sin^2 \frac{\theta}{2} \right), \quad (6.18)$$

with G_1, G_2 re-expressed in terms of G_E, G_M and kinematic factors. Empirically, a simple approximate fit is

$$G_E(Q^2) \simeq \frac{1}{1 + Q^2/Q_0^2}, \qquad G_M(Q^2) \simeq \frac{g_p}{2} \, G_E(Q^2), \qquad Q_0^2 \simeq 0.71 \text{ GeV}^2, \quad (6.19)$$

illustrating that form factors decrease with momentum transfer: larger Q^2 resolves smaller distance scales and reveals finite size.

6.3 Deep Inelastic Scattering (DIS): from form factors to structure functions

6.3.1 Kinematics and why DIS is different

Consider inclusive scattering

$$e^-(p_A) + p(p_B) \to e^-(p_1) + X,$$

where X is *any* hadronic final state. We measure only the outgoing electron. Define the momentum transfer $q = p_A - p_1$ and $Q^2 = -q^2 > 0$.

Compared with elastic scattering (where $X = p$), in DIS we must sum over all hadronic states:

$$\sum_X (2\pi)^4 \delta^{(4)}(p_A + p_B - p_1 - p_X) \langle X|j^\mu_{\rm QCD}(0)|p\rangle \langle X|j^\nu_{\rm QCD}(0)|p\rangle^*. \tag{6.20}$$

This motivates defining the **hadronic tensor** $W^{\mu\nu}$:

$$\boxed{W^{\mu\nu}(p_B, q) \equiv \frac{1}{4\pi m_p} \sum_X (2\pi)^4 \delta^{(4)}(p_B + q - p_X) \langle p|j^\mu_{\rm QCD}(0)|X\rangle \langle p|j^\nu_{\rm QCD}(0)|X\rangle^*.} \tag{6.21}$$

6.3.2 General decomposition: structure functions W_1 and W_2

Lorentz invariance, parity invariance, and current conservation imply the standard decomposition:

$$\boxed{W^{\mu\nu}(p_B, q) = W_1(Q^2, \nu)\left(-g^{\mu\nu} + \frac{q^\mu q^\nu}{q^2}\right) + W_2(Q^2, \nu)\left(p_B^\mu - \frac{p_B \cdot q}{q^2} q^\mu\right)\left(p_B^\nu - \frac{p_B \cdot q}{q^2} q^\nu\right),} \tag{6.22}$$

where the energy transfer variable is

$$\nu \equiv \frac{p_B \cdot q}{m_p} = q^0_{\rm LAB} = E_A - E_1 \quad \text{(LAB frame)}. \tag{6.23}$$

The scalar functions W_1 and W_2 are the **structure functions**. They play the role that F_1, F_2 played in elastic scattering, but now they depend on *two* Lorentz invariants (Q^2, ν) because the final hadronic state is not fixed.

6.3.3 DIS cross section in terms of W_1 and W_2

The cross section can be written as a contraction of the leptonic tensor $L_e^{\mu\nu}$ (with known QED form) and $W_{\mu\nu}$:

$$\left(\frac{d\sigma}{dE_1\, d\Omega}\right)_{\rm LAB} = \frac{4\alpha^2 E_1^2}{q^4}\left(W_2(Q^2, \nu)\cos^2\frac{\theta}{2} + 2W_1(Q^2, \nu)\sin^2\frac{\theta}{2}\right), \tag{6.24}$$

where θ is the electron scattering angle in the LAB and $q^2 \simeq -4E_A E_1 \sin^2(\theta/2)$.

6.3.4 Bjorken scaling: a surprise from experiment

For scattering off an *elementary* spin-$\frac{1}{2}$ target, kinematics plus the delta-function constraints imply that the analogs of νW_2 and $2mW_1$ depend on a *single* variable rather than on (Q^2, ν) separately. This motivates introducing the dimensionless scaling variable

$$x \equiv \frac{Q^2}{2m_p\nu}, \qquad \text{equivalently} \qquad \omega \equiv \frac{2m_p\nu}{Q^2} = \frac{1}{x}. \tag{6.25}$$

SLAC experiments found that, at large enough Q^2, the proton behaves as if composed of **quasi-free point-like constituents**: the structure functions approximately scale (i.e. depend mostly on x) rather than on Q^2 and ν independently. This is the phenomenological basis for the **parton model**.

6.4 Parton model: PDFs and the Callan–Gross relation

6.4.1 Partons as quasi-free constituents

Model the fast proton as a collection of partons i (quarks, antiquarks, and possibly other neutral constituents) carrying a fraction x of the proton momentum:

$$p_i^\mu \simeq x\, p_B^\mu, \qquad x \in [0, 1]. \tag{6.26}$$

Let $f_i(x)$ be the probability density (parton distribution function, PDF) that parton i carries momentum fraction x.

For a spin-$\frac{1}{2}$ parton, the parton-level kinematics implies contributions to the proton structure functions that localize at the kinematically allowed x. Summing incoherently over partons weighted by their squared electric charges Q_i^2 gives

$$F_2(x) \equiv \sum_i Q_i^2\, x\, f_i(x), \qquad F_1(x) \equiv \sum_i Q_i^2\, \frac{1}{2}\, f_i(x), \tag{6.27}$$

and therefore the **Callan–Gross relation**

$$\boxed{2x\, F_1(x) = F_2(x),} \tag{6.28}$$

which is characteristic of spin-$\frac{1}{2}$ constituents. (If partons had spin 0, one would find $F_1(x) = 0$.)

In the high-energy limit ($s \gg m_p^2$) the double-differential DIS cross section can be presented in terms of x and the inelasticity variable y:

$$y \equiv \frac{p_B \cdot q}{p_B \cdot p_A}, \qquad \frac{d\sigma}{dx\, dy} \simeq \frac{2\pi\alpha^2}{s\, q^4}\, F_2(x)\, \big(1 + (1-y)^2\big), \tag{6.29}$$

emphasizing that DIS essentially measures $F_2(x)$, and hence particular combinations of PDFs.

6.4.2 Flavor decomposition and valence vs sea

Assume the charged partons are quarks/antiquarks of flavors $u, d, s, \ldots$. Then

$$\frac{F_2(x)}{x} = \left(\frac{2}{3}\right)^2 \big(u(x) + \bar{u}(x)\big) + \left(\frac{1}{3}\right)^2 \big(d(x) + \bar{d}(x)\big) + \left(\frac{1}{3}\right)^2 \big(s(x) + \bar{s}(x)\big) + \cdots . \tag{6.30}$$

It is useful to separate

$$u(x) = u_v(x) + u_s(x), \qquad \bar{u}(x) = u_s(x), \tag{6.31}$$

and similarly for d, s. Here:

- ◇ **valence PDFs** $q_v(x)$ encode the net quantum numbers of the hadron,
- ◇ **sea PDFs** $q_s(x)$ arise from $q\bar{q}$ pair production.

At small x (large energy transfer), the sea dominates; at large x the valence dominates.

6.4.3 Sum rules: flavor number and momentum

PDFs satisfy powerful normalization constraints:

- ◇ **Flavor number (valence) sum rules:**

$$\int_0^1 dx\,[u(x) - \bar{u}(x)] = \int_0^1 dx\, u_v(x) = 2, \qquad \int_0^1 dx\,[d(x) - \bar{d}(x)] = \int_0^1 dx\, d_v(x) = 1, \tag{6.32}$$

  for the proton (uud).

- ◇ **Momentum sum rule:** the proton momentum is carried by all partons, including electrically neutral ones:

$$1 = \int_0^1 dx\, x\Big(u + \bar{u} + d + \bar{d} + s + \bar{s} + g + \cdots\Big), \tag{6.33}$$

  where $g(x)$ denotes the PDF of electrically neutral partons.

Experimentally, integrating measured F_2 moments indicates that a substantial fraction of the proton momentum is carried by electrically neutral partons that do not couple directly to the photon. In QCD, these are naturally identified with **gluons**.

6.4.4 Scaling violations: the footprint of QCD

Bjorken scaling is only approximate: structure functions develop a mild dependence on Q^2. In QCD this arises because quarks radiate gluons (and gluons split into $q\bar{q}$), so the parton picture evolves with the resolution scale. At very small x the Q^2 dependence becomes stronger. This is a key bridge from QED-based DIS phenomenology to QCD dynamics.

6.5 From QED phenomenology to QCD: why the strong force must be non-abelian

DIS and hadron spectroscopy suggest the underlying strong theory must include:

1. spin-$\frac{1}{2}$ quarks with **color** and **flavor**,
2. only **color singlet** physical states (confinement),
3. approximate **chiral symmetry** for light quarks,
4. **spontaneous chiral symmetry breaking** to the vector subgroup,
5. **asymptotic freedom**: quarks behave as free at large momentum transfer,
6. and **electrically neutral constituents** carrying momentum (gluons).

This points to a local non-abelian gauge theory: **QCD** with gauge group $SU(3)_{\text{color}}$.

6.5.1 Local SU(3) gauge invariance and the QCD Lagrangian

Let $q(x)$ be a color triplet:

$$q(x) = \begin{pmatrix} q_1(x) \\ q_2(x) \\ q_3(x) \end{pmatrix}, \qquad q(x) \to g(x)\, q(x), \qquad g(x) \in SU(3).$$

To make $\bar{q} i \partial\!\!\!/ q$ invariant under local $g(x)$, introduce a covariant derivative

$$D_\mu = \partial_\mu + i g_s\, G_\mu(x), \qquad G_\mu(x) = T^a G^a_\mu(x), \tag{6.34}$$

where T^a are the $SU(3)$ generators in the fundamental representation. The gauge field transforms as

$$G_\mu \to g G_\mu g^\dagger - \frac{i}{g_s}\, g\, \partial_\mu g^\dagger. \tag{6.35}$$

Define the field strength through the commutator:

$$i g_s\, G_{\mu\nu} \equiv [D_\mu, D_\nu], \qquad G_{\mu\nu} = T^a G^a_{\mu\nu}, \qquad G^a_{\mu\nu} = \partial_\mu G^a_\nu - \partial_\nu G^a_\mu - g_s f^{abc} G^b_\mu G^c_\nu, \tag{6.36}$$

where $[T^a, T^b] = i f^{abc} T^c$. The Yang–Mills term is

$$\mathcal{L}_{\text{YM}} = -\frac{1}{2}\text{tr}(G_{\mu\nu} G^{\mu\nu}) = -\frac{1}{4} G^a_{\mu\nu} G^{a\,\mu\nu}. \tag{6.37}$$

Therefore, for one flavor,

$$\mathcal{L}_{\text{QCD}} = -\frac{1}{4} G^a_{\mu\nu} G^{a\,\mu\nu} + \bar{q}\big(i D\!\!\!/ - m_q\big) q, \tag{6.38}$$

and for multiple flavors q_j one sums over $j = u, d, s, \ldots$ with their masses m_j.

Key qualitative difference from QED. In QED, photons do not carry electric charge, so there are no photon–photon self-interactions. In QCD, gluons carry color charge, so the Yang–Mills term contains **three-gluon** and **four-gluon** interactions. This non-abelian structure is crucial for asymptotic freedom and confinement.

6.6 $e^-e^+ \to$ hadrons and the number of colors

Define the ratio

$$R(e^-e^+ \to \text{hadrons}) \equiv \frac{\sigma(e^-e^+ \to \text{hadrons})}{\sigma(e^-e^+ \to \mu^+\mu^-)}. \tag{6.39}$$

At sufficiently high energies, the photon couples to quark–antiquark pairs. A common *duality* assumption is that inclusive hadron production equals inclusive parton production:

$$\sigma(e^-e^+ \to \text{hadrons}) \approx \sigma(e^-e^+ \to q\bar{q} + X),$$

and in the simplest approximation (neglecting strong interactions at high energy),

$$R \simeq \sum_{j\,\text{open}} N_c\, Q_j^2, \tag{6.40}$$

where the sum runs over quark flavors kinematically accessible, Q_j is the quark charge in units of e, and N_c is the number of colors. Comparing to data gives strong evidence for $N_c = 3$.

6.7 Heavy quarks: NRQCD, HQET, and quarkonium

For heavy quarks (charm, bottom), $m_Q \gg \Lambda_{\text{QCD}}$:

- ◇ the strong coupling at the scale m_Q is relatively small,
- ◇ heavy quarks in hadrons move non-relativistically ($v \ll 1$),
- ◇ effective theories become powerful: NRQCD and HQET.

6.7.1 NRQCD: non-relativistic QCD for heavy quarks

A heavy quark can be described by a Schrödinger field ψ in the color triplet, with an expansion in $1/m_Q$:

$$\mathcal{L}_{\text{NRQCD}} = \psi^\dagger \left(iD_0 + \frac{\mathbf{D}^2}{2m_Q} + \mu \cdot \mathbf{B} + \cdots \right) \psi, \qquad \mu \sim \frac{g_s}{m_Q}\,\mathbf{S}, \tag{6.41}$$

where $\mathbf{B}$ is the chromomagnetic field (a matrix in color space). The chromomagnetic term controls spin splittings.

6.7.2 HQET: heavy–light hadrons and heavy quark symmetry

For heavy–light hadrons $H = (Q\,l)$ (one heavy quark plus light degrees of freedom), the hadron's center of mass is close to the heavy quark position, so the heavy quark acts approximately as a static color source. At leading order,

$$\mathcal{L}_{\text{HQET}} \simeq \psi^\dagger i D_0 \psi. \tag{6.42}$$

This yields two approximate symmetries:

- ◇ **Flavor symmetry:** leading dynamics independent of m_Q,
- ◇ **Spin symmetry:** heavy-quark spin decouples at leading order.

Consequently, heavy–light hadron masses admit an expansion

$$M_H = m_Q + \Lambda_0^{(l)} + \mathcal{O}\left(\frac{1}{m_Q}\right), \tag{6.43}$$

and spin partners are approximately degenerate with splitting

$$M_{H^*} - M_H = \frac{\Lambda_2^{(l)}}{m_Q} + \mathcal{O}\left(\frac{1}{m_Q^2}\right). \tag{6.44}$$

This explains why splittings decrease for heavier quarks (bottom vs charm).

6.7.3 Heavy quarkonium: bound states $(Q\bar{Q})$ and gluonic decays

For quarkonium $H = (Q\bar{Q})$, both constituents are heavy and move non-relativistically about the center of mass. A flavor-independent potential picture is often reasonable, analogous to positronium, but with gluons instead of photons.

Decays to *light hadrons* proceed via annihilation into gluons. Since QCD is invariant under parity and charge conjugation, the number of gluons in the dominant decay depends on the state's J^{PC} and the gluon quantum numbers (gluons have $J^{PC} = 1^{--}$ for the field). A key selection rule is:

- ◇ states with J^{P+} can decay to **two** gluons,
- ◇ states with J^{P-} typically require **three** gluons.

Because each extra gluon brings a factor of $\alpha_s(m_Q)$, three-gluon decays are parametrically suppressed relative to two-gluon decays:

$$\frac{\Gamma\big((Q\bar{Q})[J^{P-}] \to ggg\big)}{\Gamma\big((Q\bar{Q})[J^{P+}] \to gg\big)} \sim \alpha_s(m_Q) \ll 1. \tag{6.45}$$

For the ground states, ${}^1S_0$ has $J^{PC} = 0^{-+}$ and ${}^3S_1$ has $J^{PC} = 1^{--}$, so the vector state requires three gluons while the pseudoscalar can go to two gluons, leading to strong hierarchies of hadronic widths (up to additional numerical factors).

6.8 Summary and conceptual map

- **Exclusive processes** (fixed hadronic final state) are described by **form factors**, obtained by the most general Lorentz-covariant decomposition of current matrix elements plus symmetry constraints.
- **Elastic scattering on spin-$\frac{1}{2}$ hadrons** needs two form factors: $F_1(q^2)$ and $F_2(q^2)$ (or equivalently G_E and G_M). The existence of large $F_2(0)$ reflects intrinsic structure (anomalous magnetic moment).
- **Inclusive processes** like DIS require summing over hadronic final states, leading to the hadronic tensor $W^{\mu\nu}$ and **structure functions** W_1, W_2.
- **Bjorken scaling** and the **Callan–Gross relation** provide evidence that the proton contains **spin-$\frac{1}{2}$ partons**.
- **PDF sum rules** plus DIS moments imply the presence of **electrically neutral partons** carrying momentum: naturally identified with **gluons**.
- These facts motivate **QCD** as a non-abelian $SU(3)$ gauge theory with gluon self-interactions, explaining (qualitatively) both asymptotic freedom and the need for color singlets.
- For **heavy quarks**, effective theories (NRQCD/HQET) yield approximate spin/flavor symmetries and explain observed degeneracy patterns and decay hierarchies.

7 . Weak Interactions

7.1 Why do we need weak interactions? (a first phenomenological picture)

A striking experimental fact is that not all unstable hadrons decay with comparable widths. Typical widths of hadron resonances (like ρ, ω, Δ) are of order

$$\Gamma_{\text{strong}} \sim 1\text{–}100 \text{ MeV}. \tag{7.1}$$

However, some of the *lightest* hadrons in the lowest isospin multiplets (such as $\pi^{\pm}, \pi^0, p, n$) have decay widths that are *many orders of magnitude smaller*. The qualitative message is:

- ◇ Isospin is (approximately) conserved by the **strong** interactions.
- ◇ Therefore, the lightest hadrons in isospin multiplets must decay through **other** interactions.

Two classic examples:

- ◇ π^0 decays electromagnetically with a width of order $\Gamma_{\text{em}} \sim 10^{-5}$ MeV.
- ◇ $\pi^{\pm}$ and the neutron decay weakly, with much smaller widths $\Gamma_{\text{weak}} \sim 10^{-14}$–$10^{-24}$ MeV.

Weak interactions are also required to explain quantitatively:

- ◇ Muon decay:

$$\Gamma_{\mu^- \to e^- \bar{\nu}_e \nu_\mu} \;=\; 2.6 \times 10^{-16} \text{ MeV}. \tag{7.2}$$

- ◇ Decays of the lightest strange hadrons (example numerical widths):

$$\Gamma_{K^{\pm}} = 4.6 \times 10^{-16} \text{ MeV}, \qquad \Gamma_{K^0_S} = 6.3 \times 10^{-12} \text{ MeV}, \tag{7.3}$$

$$\Gamma_{K^0_L} = 1.1 \times 10^{-14} \text{ MeV}, \qquad \Gamma_{\Lambda^0} = 2.1 \times 10^{-12} \text{ MeV}. \tag{7.4}$$

7.2 Historical origin: beta decay and the neutrino

Weak interactions were historically discovered in nuclear beta decay:

$$^{A}_{Z}X \to {}^{A}_{Z+1}X + e^- \qquad (\beta^- \text{ decay}), \tag{7.5}$$

$$^{A}_{Z}X \to {}^{A}_{Z-1}X + e^+ \qquad (\beta^+ \text{ decay}). \tag{7.6}$$

In these decays, the observed electron (or positron) energy spectrum is continuous. To explain the missing energy, momentum, and angular momentum, Pauli postulated the existence of the neutrino ν: a spin-$\frac{1}{2}$ particle, originally taken to be massless.

At the nucleon level, beta decay can be understood in terms of

$$n \to p\, e^- \, \bar{\nu}_e, \qquad p \to n\, e^+ \, \nu_e. \tag{7.7}$$

7.2.1 Fermi theory (parity invariant version)

The first interaction proposed by Fermi was a local four-fermion interaction of the form

$$\mathcal{L}_{\text{int}} = G\, \bar{\psi}_p \gamma^\mu \psi_n \; \bar{\psi}_e \gamma_\mu \psi_{\nu_e} + \text{H.c.} \equiv G\, \bar{p} \gamma^\mu n \; \bar{e} \gamma_\mu \nu_e + \text{H.c.} \tag{7.8}$$

This interaction is **parity invariant**.

However, a parity invariant weak theory runs into difficulties when trying to accommodate the observed decay pattern of the kaon. Indeed, for K^+ decays:

- From $K^+ \to \pi^+ \pi^0$ one would infer $P(K^+) = +$.
- From $K^+ \to \pi^+ \pi^0 \pi^0$ one would infer $P(K^+) = -$.

This tension points to the need for **parity violation** in weak interactions.

7.3 Parity violation and the V–A structure

7.3.1 Lee–Yang proposal and Wu experiment

Lee and Yang proposed (1956) a direct test of parity violation in beta decay using polarized nuclei. The idea: polarize ^{60}Co and study the angular distribution of emitted electrons in

$$^{60}\text{Co} \to {}^{60}\text{Ni}^* \, e^- \, \bar{\nu}_e. \tag{7.9}$$

The magnetic field used to polarize the nucleus is invariant under parity. Therefore:

- If parity were conserved, the number of electrons emitted along and opposite to the polarization direction would be the same.
- Wu's experiment (1956) observed a clear asymmetry: electrons preferentially go opposite to the polarization direction.

This suggests a chiral structure: **only the left-handed component of leptons is sensitive to weak interactions.**

7.3.2 V–A (vector minus axial) Fermi interaction

Fermi's original proposal was modified to the **V–A** form

$$\boxed{\mathcal{L}_{\text{int}} = 2\sqrt{2}\, G\, \bar{p}\, \gamma^\mu P_L\, n\; \bar{e}\, \gamma_\mu P_L\, \nu_e + \text{H.c.} \qquad P_L = \frac{1-\gamma^5}{2}} \tag{7.10}$$

Key qualitative consequences:

- ◇ If the outgoing charged lepton mass is negligible, the charged lepton is produced *fully polarized*.
- ◇ All fields are evaluated at the same spacetime point ⇒ it is a **zero-range** interaction.
- ◇ The coupling has mass dimension $[G] = -2$, suggesting an **effective** interaction arising as a low-energy limit of a more fundamental finite-range theory.

7.3.3 Finite-range origin: massive vector boson mediator

A zero-range interaction can emerge as the low-energy limit of a massive vector boson exchange. If W_μ is a massive vector field with mass m_W, then its propagator schematically is

$$\langle 0|T\{W_\mu(x)W_\nu^\dagger(y)\}|0\rangle = \int \frac{d^4k}{(2\pi)^4} \frac{i\, e^{-ik\cdot(x-y)}}{k^2 - m_W^2} \left(-g_{\mu\nu} + \frac{k_\mu k_\nu}{m_W^2}\right). \tag{7.11}$$

In the low-momentum regime $k_\mu \ll m_W$,

$$\frac{1}{k^2 - m_W^2} \simeq -\frac{1}{m_W^2}, \tag{7.12}$$

$$\Rightarrow \quad \langle 0|T\{W_\mu(x)W_\nu^\dagger(y)\}|0\rangle \simeq \int \frac{d^4k}{(2\pi)^4}\, e^{-ik\cdot(x-y)}\, \frac{ig_{\mu\nu}}{m_W^2} = \frac{ig_{\mu\nu}}{m_W^2}\, \delta^{(4)}(x-y). \tag{7.13}$$

Hence one expects parametrically

$$\boxed{G \sim \frac{1}{m_W^2}.} \tag{7.14}$$

7.4 Fermi theory at the quark level and global symmetries

Since $n = (udd)$ and $p = (uud)$, beta decay is naturally interpreted as a quark-level transition

$$d \to u\, e^-\, \bar{\nu}_e. \tag{7.15}$$

The Fermi theory can be written in terms of quark and lepton fields as

$$\boxed{\mathcal{L}_{\text{int}} = 2\sqrt{2}\, G\, J^\mu J_\mu^\dagger, \qquad J^\mu = \bar{u}\gamma^\mu P_L d + \bar{\nu}_e \gamma^\mu P_L e + \bar{\nu}_\mu \gamma^\mu P_L \mu + \cdots} \tag{7.16}$$

Immediate consequences emphasized by the theory:

- ◇ If a right-handed neutrino exists, it does not interact in this framework (a **sterile** neutrino).
- ◇ The interaction is invariant under independent global phase rotations of each left-handed lepton doublet:

$$\ell_L \to e^{i\theta_\ell} \ell_L, \qquad \nu_{\ell L} \to e^{i\theta_\ell} \nu_{\ell L}, \qquad \ell = e, \mu, \ldots \tag{7.17}$$

- ◇ With the corresponding transformation of right-handed charged leptons $\ell_R \to e^{i\theta_\ell}\ell_R$, the full Lagrangian remains invariant for constant $\theta_\ell \Rightarrow$ individual leptonic numbers (electronic, muonic, ...) are conserved.
- ◇ Similarly, invariance under a global baryon rotation $q \to e^{i\theta_B} q$ (constant θ_B) implies **baryon number conservation**.

A key question for low-energy processes is: if the weak interaction is written at the quark level, how should it be represented at the **hadronic** level, where nucleons and pions are the degrees of freedom? The answer involves symmetry constraints (chiral symmetry) and, in general, form factors. We will see that chiral symmetry strongly constrains the hadronic structure of weak interactions.

7.5 Low-energy test I: charged pion decay and helicity suppression

7.5.1 Process and relevant interaction

Consider the charged pion decay

$$\pi^+ \to \ell^+ \nu_\ell, \qquad \ell = \mu, e. \tag{7.18}$$

The relevant piece of the Fermi Lagrangian is

$$\mathcal{L}_{\text{int}} = 2\sqrt{2}\, G\, J_q^{\mu\dagger} J_{\ell\mu}, \qquad J_q^{\mu\dagger} = \bar{d}\gamma^\mu P_L u, \qquad J_\ell^\mu = \bar{\nu}_\ell \gamma^\mu P_L \ell. \tag{7.19}$$

At first order,

$$\mathcal{M} = \langle f|\mathcal{L}_{\text{int}}(0)|i\rangle = 2\sqrt{2}\, G\, \underbrace{\langle 0|\bar{d}\gamma^\mu P_L u|\pi^+(p_A)\rangle}_{\text{hadronic}}\, \underbrace{\langle \ell^+(p_1,\lambda_1)\, \nu_\ell(p_2,\lambda_2)|\bar{\nu}_\ell \gamma_\mu P_L \ell|0\rangle}_{\text{leptonic}}. \tag{7.20}$$

Important helicity observation:

- ◇ The outgoing neutrino is left-handed $\Rightarrow$ its helicity is fixed: $\lambda_2 = -1$.
- ◇ If the charged lepton mass were negligible, the outgoing ℓ^+ would be produced with fixed helicity $\lambda_1 = +1$ (fully polarized).
- ◇ For a pion at rest ($J^P = 0^-$), angular momentum conservation would then fail: the amplitude must be proportional to m_ℓ.

This is the origin of **helicity suppression**.

7.5.2 Leptonic matrix element

For the leptonic part,

$$\langle \ell^+(p_1,\lambda_1)\, \nu_\ell(p_2,\lambda_2)|\bar{\nu}_\ell \gamma^\mu P_L \ell|0\rangle = \bar{u}(2)\gamma^\mu P_L v(1). \tag{7.21}$$

7.5.3 Hadronic matrix element and the pion decay constant

For the hadronic part, define

$$\langle 0|\bar{d}\gamma^\mu P_L u|\pi^+(p_A)\rangle \equiv \frac{i}{\sqrt{2}}\, F_\pi\, p_A^\mu. \tag{7.22}$$

Since the pion has $J^P = 0^-$, parity implies that only the axial part contributes:

$$\bar{d}\gamma^\mu P_L u = \frac{1}{2}\,\bar{d}\gamma^\mu u - \frac{1}{2}\,\bar{d}\gamma^\mu\gamma^5 u, \qquad \Rightarrow \qquad \langle 0|\bar{d}\gamma^\mu u|\pi^+\rangle = 0, \tag{7.23}$$

and

$$\langle 0|\bar{d}\gamma^\mu P_L u|\pi^+\rangle = -\frac{1}{2}\,\langle 0|\bar{d}\gamma^\mu\gamma^5 u|\pi^+\rangle. \tag{7.24}$$

The constant F_π is the (weak) **pion decay constant**.

7.5.4 Amplitude and explicit m_ℓ factor

Putting both parts together,

$$\mathcal{M} = 2\sqrt{2}\,G\left(\frac{i}{\sqrt{2}}F_\pi p_A^\mu\right)\bar{u}(2)\gamma_\mu P_L v(1) = 2iGF_\pi\, p_A^\mu\,\bar{u}(2)\gamma_\mu P_L v(1). \tag{7.25}$$

Now use $p_A = p_1 + p_2$ and the spinor equations of motion (as used in the slides):

$$\bar{u}(2)\not{p}_2 = 0, \tag{7.26}$$

$$(\not{p}_1 + m_\ell)\, v(1) = 0, \tag{7.27}$$

so that

$$\bar{u}(2)\not{p}_A P_L v(1) = \bar{u}(2)(\not{p}_1 + \not{p}_2)P_L v(1) = \bar{u}(2)\not{p}_1 P_L v(1) = -m_\ell\,\bar{u}(2)P_R v(1). \tag{7.28}$$

Therefore,

$$\boxed{\mathcal{M} = -2iGF_\pi\, m_\ell\,\bar{u}(2)P_R v(1)} \tag{7.29}$$

This makes helicity suppression manifest: $\mathcal{M} \propto m_\ell$.

7.5.5 Polarization selection and decay width

In the pion rest frame, the slides compute explicitly

$$\mathcal{M}\big(\pi^+(p_A = 0) \to \ell^+(p_1, +)\,\nu_\ell(p_2, -)\big) = 0, \tag{7.30}$$

$$\mathcal{M}\big(\pi^+(p_A = 0) \to \ell^+(p_1, -)\,\nu_\ell(p_2, -)\big) \neq 0, \tag{7.31}$$

i.e. the charged lepton is fully polarized in this approximation.

Summing over helicities (as done in the slides) yields

$$\sum_{\lambda_1=\pm} |\mathcal{M}|^2 = 8G^2 F_\pi^2 m_\ell^2\, E_2\, m_\pi, \tag{7.32}$$

with $p = E_2 = \sqrt{E_1^2 - m_\ell^2}$ and $E_1 + E_2 = m_\pi$ in the pion rest frame. Using

$$p = \frac{m_\pi^2 - m_\ell^2}{2m_\pi}, \tag{7.33}$$

the decay width becomes

$$\boxed{\Gamma(\pi^+ \to \ell^+\nu_\ell) = \frac{G^2 F_\pi^2 m_\ell^2 m_\pi}{4\pi}\left(1 - \frac{m_\ell^2}{m_\pi^2}\right)^2} \tag{7.34}$$

7.5.6 A parameter-free prediction: the e/μ ratio

Taking the ratio cancels G and F_π:

$$\boxed{\frac{\Gamma(\pi^+ \to e^+\nu_e)}{\Gamma(\pi^+ \to \mu^+\nu_\mu)} = \frac{m_e^2}{m_\mu^2}\left(\frac{m_\pi^2 - m_e^2}{m_\pi^2 - m_\mu^2}\right)^2 \simeq 1.2 \times 10^{-4},} \tag{7.35}$$

in agreement with the quoted experimental value $\simeq 1.2 \times 10^{-4}$. This is a clean low-energy confirmation of the V–A structure.

7.6 Non-linear sigma model: implementing weak interactions at hadronic level

Pion decay and beta decay are low-energy processes, so they should be describable within the non-linear sigma model (effective hadronic theory). The guiding principle is to **implement the symmetry breaking pattern**.

7.6.1 External left-handed field and local chiral symmetry

Start from the quark-level charged current:

$$\mathcal{L}_{\text{int}} = 2\sqrt{2}\, G\, J_q^{\mu\dagger} J_{\ell\mu} + \text{H.c.}, \qquad J_q^{\mu\dagger} = \bar{d}\gamma^\mu P_L u. \tag{7.36}$$

Introduce the isospin doublet $q = (u, d)^T$ and

$$\tau^+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad \tau^- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \tag{7.37}$$

so

$$J_q^{\mu\dagger} = \bar{q}\gamma^\mu P_L \tau^- q, \qquad J_q^\mu = \bar{q}\gamma^\mu P_L \tau^+ q. \tag{7.38}$$

Then one can write

$$\mathcal{L}_{\text{int}} = \bar{q}\,\gamma^\mu P_L\, a_\mu^L\, q, \tag{7.39}$$

with an *external* matrix field

$$a_\mu^L \equiv 2\sqrt{2}\,G \begin{pmatrix} 0 & \bar{\ell}\gamma_\mu P_L \nu_\ell \\ \bar{\nu}_\ell \gamma_\mu P_L \ell & 0 \end{pmatrix}. \tag{7.40}$$

In the massless limit $m_u = m_d = 0$, the QCD Lagrangian has a global $SU_L(2) \otimes SU_R(2)$ chiral symmetry. The term above breaks it because $[g_L, a_\mu^L] \neq 0$ for constant $g_L \in SU(2)$. The trick is:

- ◇ Promote g_L to a **local** transformation $g_L(x)$ and assign the transformation law

$$a_\mu^L(x) \to g_L(x) a_\mu^L(x) g_L^\dagger(x) + i g_L(x) \partial_\mu g_L^\dagger(x), \qquad g_R = \text{constant}. \tag{7.41}$$

- ◇ Then the left-handed quark kinetic term becomes locally invariant if we replace ∂_μ by $(\partial_\mu - i a_\mu^L)$ on q_L:

$$\mathcal{L} = \bar{q}_L (i\partial\!\!\!/ + a\!\!\!/_L) q_L + \bar{q}_R i\partial\!\!\!/ q_R, \tag{7.42}$$

  and for QCD the same holds with $\partial \to D$ (including gluons).

7.6.2 Coupling to the non-linear sigma model

In the non-linear sigma model, the pion field is encoded in

$$U(x) \in SU(2), \qquad U(x) \to g_L(x) U(x) g_R^\dagger. \tag{7.43}$$

Because g_L is local, $\partial_\mu U$ does not transform covariantly. Define the covariant derivative

$$D_\mu U \equiv (\partial_\mu - i a_\mu^L)\, U, \qquad D_\mu U \to g_L(x) (D_\mu U) g_R^\dagger, \tag{7.44}$$

and replace

$$\mathcal{L} = \frac{f_\pi^2}{4} \operatorname{tr}(\partial_\mu U^\dagger \partial^\mu U) \quad \longrightarrow \quad \mathcal{L} = \frac{f_\pi^2}{4} \operatorname{tr}\big[(D_\mu U)^\dagger D^\mu U\big]. \tag{7.45}$$

Expanding in G gives the weak interaction term (to $\mathcal{O}(G)$):

$$\mathcal{L}_{\text{int}}^{\text{weak}} = \frac{f_\pi^2}{4} \operatorname{tr}\Big(\partial_\mu U^\dagger (-i a_L^\mu) U + U^\dagger i a_L^\mu \partial_\mu U\Big) + \mathcal{O}(G^2) \tag{7.46}$$

$$\simeq i\, 2\sqrt{2}\, G\, \frac{f_\pi^2}{4} \operatorname{tr}\left[\begin{pmatrix} 0 & \bar{\ell}\gamma_\mu P_L \nu_\ell \\ \bar{\nu}_\ell \gamma_\mu P_L \ell & 0 \end{pmatrix} \left(\partial^\mu U\, U^\dagger - U \partial^\mu U^\dagger\right)\right]. \tag{7.47}$$

To leading order in pion fields,

$$\boxed{\mathcal{L}_{\text{int}}^{\text{weak}} \simeq -2G f_\pi \left(\partial_\mu \pi^+\, \bar{\nu}_\ell \gamma^\mu P_L \ell + \partial_\mu \pi^-\, \bar{\ell} \gamma^\mu P_L \nu_\ell\right).} \tag{7.48}$$

Comparing with the quark-level interaction identifies the hadronic representation of the current:

$$J_q^{\mu\dagger} \;\to\; \frac{i f_\pi^2}{4} \operatorname{tr}\big[\tau^- \left(\partial^\mu U\, U^\dagger - U \partial^\mu U^\dagger\right)\big] \simeq -\frac{f_\pi}{\sqrt{2}}\, \partial^\mu \pi^+ + \cdots \tag{7.49}$$

7.6.3 Recovering pion decay and the relation $F_\pi = f_\pi$

Using the hadronic current above,

$$\langle 0|\bar{d}\gamma^\mu P_L u|\pi^+(p_A)\rangle \;\parallel\; \left\langle 0 \left| -\frac{f_\pi}{\sqrt{2}}\,\partial^\mu \pi^+ \right| \pi^+(p_A) \right\rangle = \frac{i f_\pi}{\sqrt{2}}\, p_A^\mu, \tag{7.50}$$

so we conclude

$$\boxed{F_\pi = f_\pi.} \tag{7.51}$$

This is a powerful EFT statement: f_π not only controls low-energy **strong** pion interactions, it also fixes the normalization of low-energy **weak** pion processes. Moreover, at low energy one can compute other weak pion processes (e.g. $\pi^+ \to \pi^0 e^+ \nu_e$) by expanding the same current to higher order in pion fields, without introducing new parameters.

7.7 Weak interactions with nucleons and the axial coupling g_A

For nucleons, the non-linear sigma model (as used in the slides) contains

$$\bar{N}\left(i\partial\!\!\!/ + v\!\!\!/ + i g_A a\!\!\!/\gamma^5 - m_N\right) N, \qquad v_\mu = \frac{1}{2}\left(u\partial_\mu u^\dagger + u^\dagger \partial_\mu u\right), \quad a_\mu = \frac{1}{2}\left(u\partial_\mu u^\dagger - u^\dagger\partial_\mu u\right), \quad U = u^2. \tag{7.52}$$

To incorporate the weak external field $a_\mu^L(x)$ and local $SU_L(2)$, one replaces

$$\partial_\mu u \to D_\mu u = (\partial_\mu - i a_\mu^L) u, \tag{7.53}$$

and correspondingly

$$v_\mu \to v'_\mu \equiv \frac{1}{2}\left(u\partial_\mu u^\dagger + u^\dagger D_\mu u\right), \qquad a_\mu \to a'_\mu \equiv \frac{1}{2}\left(u\partial_\mu u^\dagger - u^\dagger D_\mu u\right). \tag{7.54}$$

This yields, at $\mathcal{O}(G)$, the nucleon weak interaction term

$$\mathcal{L}_{\text{int}}^{\text{weak}} = \frac{1}{2}\,\bar{N}\left(u^\dagger a\!\!\!/_L u - g_A\, u^\dagger a\!\!\!/_L u\,\gamma^5\right) N. \tag{7.55}$$

7.7.1 Hadronic charged current at nucleon level

Comparing with the quark-level interaction gives the hadronic representation

$$J_q^{\mu\dagger} \;\to\; \frac{1}{2}\,\bar{N}\gamma^\mu\left(u^\dagger\tau^- u - g_A\, u^\dagger \tau^- u\,\gamma^5\right) N \simeq \bar{n}\,\gamma^\mu \frac{1 - g_A\gamma^5}{2}\, p + \cdots \tag{7.56}$$

For processes with *no pions* (like ordinary beta decay) we set $u = 1$ and obtain

$$\boxed{\mathcal{L}_{\text{int}}^{\text{weak}} = 2\sqrt{2}\,G\,\bar{n}\,\gamma^\mu \frac{1 - g_A\gamma^5}{2}\, p\;\bar{\nu}_\ell \gamma_\mu P_L \ell + \text{H.c.}} \tag{7.57}$$

A crucial point emphasized in the slides:

$$g_A \simeq 1.27 \quad \Rightarrow \quad \text{at nucleon level the weak interaction is } \textit{not purely left-handed}. \tag{7.58}$$

Indeed,

$$\frac{1 - g_A\gamma^5}{2} = \frac{1 + g_A}{2}\, P_L + \frac{1 - g_A}{2}\, P_R \approx 1.13\, P_L + 0.13\, P_R. \tag{7.59}$$

7.8 Nuclear beta decay as a low-energy limit

Nuclei are bound states of non-relativistic nucleons. Hence nuclear beta decay is a low-energy process, and we can start from the nucleon-level interaction and further simplify it using non-relativistic (Schrödinger) fields.

7.8.1 Non-relativistic reduction and Fermi vs. Gamow–Teller

For $|\mathbf{p}| \ll m$,

$$u_\lambda(\mathbf{p}) = \sqrt{E+m} \begin{pmatrix} \chi_\lambda \\ \frac{\sigma\cdot\mathbf{p}}{E+m}\chi_\lambda \end{pmatrix} \simeq \sqrt{2m} \begin{pmatrix} \chi_\lambda \\ 0 \end{pmatrix}. \tag{7.60}$$

Write the nucleon field as

$$N(x) = e^{-imx^0} P_+\, \phi_N(x), \qquad P_+ = \frac{1+\gamma^0}{2} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \tag{7.61}$$

where $\phi_N(x)$ is a Schrödinger spin-$\frac{1}{2}$ field. Using the projector identities quoted:

$$P_+\gamma^\mu P_+ = (1, \mathbf{0}), \qquad P_+\gamma^\mu\gamma^5 P_+ = (0, \sigma), \tag{7.62}$$

the interaction becomes

$$\boxed{\mathcal{L}_{\text{int}}^{\text{weak}} = \sqrt{2}\, G \left(\phi_n^\dagger \phi_p\, \bar{\nu}_\ell \gamma^0 P_L \ell - g_A\, \phi_n^\dagger \sigma^i \phi_p\, \bar{\nu}_\ell \gamma^i P_L \ell\right) e^{-i(m_p - m_n)x^0} + \text{H.c.}} \tag{7.63}$$

Interpretation:

- The spin-independent term $\phi_n^\dagger \phi_p$ induces **Fermi transitions**.
- The spin-dependent term $\phi_n^\dagger \sigma^i \phi_p$ induces **Gamow–Teller transitions**.
- If initial and final nuclear states have $J^P = 0^+$, the Gamow–Teller term does not contribute, so the decay width does not depend on g_A.

7.8.2 Isospin matrix elements and accurate determination of G

If, in addition, the initial and final nuclear states belong to the same isospin multiplet, then the form factor

$$\langle {}^A_{Z-1}X | \phi_n^\dagger(0)\phi_p(0) | {}^A_Z X \rangle \tag{7.64}$$

can be evaluated exactly, because $\phi_n^\dagger \phi_p$ replaces a proton by a neutron and plays the role of a τ^- operator.

Example: ${}^{14}_{6}\mathrm{C}$, ${}^{14}_{7}\mathrm{N}^*$, ${}^{14}_{8}\mathrm{O}$ have $J^P = 0^+$ and form an isospin triplet. For ${}^{14}_{8}\mathrm{O} \to {}^{14}_{7}\mathrm{N}^*\, e^+\, \nu_e$,

$$\langle {}^{14}_{7}\mathrm{N}^* | \phi_n^\dagger(0)\phi_p(0) | {}^{14}_{8}\mathrm{O}\rangle \to \langle I I_3' | \tau^- | I I_3\rangle = \sqrt{2}. \tag{7.65}$$

In such cases, G can be measured accurately:

$$\boxed{G = 1.136(3) \times 10^{-5}\ \mathrm{GeV}^{-2} \qquad (\beta\text{-decay determination}).} \tag{7.66}$$

7.9 Example calculation: spectrum and total width in nuclear beta decay

Consider (as in the slides)

$${}^{14}_{8}\mathrm{O}_A \to {}^{14}_{7}\mathrm{N}^*_1 + e^+_2 + \nu_{e3}. \tag{7.67}$$

Using the Fermi term only, the amplitude is

$$\mathcal{M} = \sqrt{2}\, G\, \langle 1 | \phi_n^\dagger \phi_p | A\rangle\, \langle 2, 3 | \bar{\nu}_\ell \gamma^0 P_L \ell | 0\rangle. \tag{7.68}$$

Inserting the isospin matrix element $\sqrt{2}$ and the relativistic normalizations used in the slides,

$$\mathcal{M} = \sqrt{2}\, G \left(\sqrt{2}\right) \left(\sqrt{2m_A}\sqrt{2m_1}\right) \bar{u}(3)\gamma^0 P_L v(2). \tag{7.69}$$

If the positron polarization is not measured, one obtains (trace steps as in the slides)

$$|\mathcal{M}|^2 = 16 G^2 m_A m_1 \sum_{\lambda_2,\lambda_3} \mathrm{tr}\Big[\not{p}_3 \gamma^0 P_L (\not{p}_2 - m_2)\gamma^0 P_L\Big] \tag{7.70}$$

$$= 32 G^2 m_A m_1 \left(E_2 E_3 + |\mathbf{p}_2||\mathbf{p}_3| \cos\theta\right), \qquad \hat{p}_2 \cdot \hat{p}_3 \equiv \cos\theta. \tag{7.71}$$

7.9.1 Phase space and spectrum (massless neutrino, m_e neglected)

The decay width is

$$\Gamma = \frac{1}{2m_A} \int \frac{d^3\mathbf{p}_1}{(2\pi)^3 2E_1} \int \frac{d^3\mathbf{p}_2}{(2\pi)^3 2E_2} \int \frac{d^3\mathbf{p}_3}{(2\pi)^3 2E_3}\, |\mathcal{M}|^2\, (2\pi)^4 \delta(m_A - E_1 - E_2 - E_3)\delta^{(3)}(\mathbf{p}_1 + \mathbf{p}_2 + \mathbf{p}_3). \tag{7.72}$$

Carrying out the integrals as done in the slides and using $E_1 \simeq m_1$, one arrives at

$$\frac{d\Gamma}{dE_2} = \frac{G^2}{\pi^3} |\mathbf{p}_2| E_2 \left(E_0 - E_2\right)^2 \simeq \frac{G^2}{\pi^3} E_2^2 \left(E_0 - E_2\right)^2 \qquad (E_2 \gg m_e), \tag{7.73}$$

where

$$E_0 \equiv m_A - m_1. \tag{7.74}$$

The total width becomes

$$\boxed{\Gamma \simeq \int_0^{E_0} dE_2\, \frac{G^2}{\pi^3} E_2^2 (E_0 - E_2)^2 = \frac{G^2 E_0^5}{30\pi^3}.} \tag{7.75}$$

7.9.2 Remarks (as emphasized in the slides)

- The width is extremely sensitive to the nuclear mass difference:

$$\Gamma \sim E_0^5 = (m_A - m_1)^5. \tag{7.76}$$

- The energy spectrum is symmetric around $E_2 \simeq E_0/2$ and peaks there.
- The electron/positron mass is not always negligible (especially near the endpoint).
- Coulomb interaction between the charged lepton and the nucleus affects fine details of the spectrum and the accurate decay width.

7.10 Finite neutrino mass effects on beta decay endpoint

Neutrinos are massless in the Standard Model, but oscillations imply a small neutrino mass. If a finite neutrino mass m_ν is included:

- The amplitude remains exactly the same.
- The phase space is modified. The decay width involves

$$\sqrt{E_3^2 - m_\nu^2}, \qquad E_3 = E_0 - E_2, \tag{7.77}$$

  and the endpoint shifts to $E_2^{\max} = E_0 - m_\nu$.

In the approximation used in the slides ($E_2 \gg m_e$, $E_0 \gg m_e$),

$$\boxed{\frac{d\Gamma}{dE_2} \simeq \frac{G^2}{\pi^3} E_2^2 \, (E_0 - E_2) \sqrt{(E_0 - E_2)^2 - m_\nu^2}.} \tag{7.78}$$

Near the endpoint, $E_2 = E_0 - m_\nu - \delta E_2$ with $\delta E_2 \to 0$,

$$\frac{d\Gamma}{dE_2} \simeq \frac{G^2}{\pi^3} E_0^2 \, m_\nu \sqrt{2 m_\nu \, \delta E_2}, \tag{7.79}$$

which highlights how endpoint measurements are sensitive to m_ν.

7.11 Muon decay: determination of G and comparison with beta decay

Consider

$$\mu_A^- \to e_1^- \, \bar{\nu}_{e2} \, \nu_{\mu 3}. \tag{7.80}$$

Since $m_\mu \simeq 106$ MeV $\gg m_e \simeq 0.5$ MeV, neglect m_e. The relevant interaction is

$$\mathcal{L}_{\text{int}} = 2\sqrt{2}\, G \; J_e^{\mu\dagger} J_{\mu\mu} + \text{H.c.}, \qquad J_e^{\mu\dagger} = \bar{e}\gamma^\mu P_L \nu_e, \qquad J_\mu^\mu = \bar{\nu}_\mu \gamma^\mu P_L \mu. \tag{7.81}$$

The amplitude is

$$\mathcal{M} = 2\sqrt{2}\, G\, \bar{u}(3)\gamma^\mu P_L u(A)\, \bar{u}(1)\gamma_\mu P_L v(2). \tag{7.82}$$

If the muon is unpolarized and final polarizations are not measured, the slides show

$$|\mathcal{M}|^2 = 4G^2\, \mathrm{tr}\Big(\not{p}_3 \gamma^\mu P_L(\not{p}_A + m_A)\gamma^\nu P_L\Big)\, \mathrm{tr}\Big(\not{p}_1 \gamma_\mu P_L \not{p}_2 \gamma_\nu P_L\Big) \tag{7.83}$$

$$\equiv 4G^2\, L^{\mu\nu}(p_3, p_A)\, L_{\mu\nu}(p_1, p_2) = 64G^2\, (p_A\cdot p_2)(p_3\cdot p_1). \tag{7.84}$$

Using $(p_A\cdot p_2) = m_A E_2$ and $(p_3\cdot p_1) = \frac{1}{2}(p_A - p_2)^2 = \frac{m_A^2}{2} - m_A E_2$, the amplitude depends only on energies (no angles), simplifying the phase-space integration.

The electron energy spectrum (with $m_A = m_\mu$) is

$$\boxed{\frac{d\Gamma}{dE_1} = \frac{G^2 m_\mu^2 E_1^2}{4\pi^3}\left(1 - \frac{4E_1}{3m_\mu}\right).} \tag{7.85}$$

The total muon decay width is

$$\boxed{\Gamma_\mu = \frac{G^2 m_\mu^5}{192\pi^3}.} \tag{7.86}$$

This allows an independent measurement:

$$G\Big|_{\mu\text{-decay}} = 1.16632(2) \times 10^{-5}\ \mathrm{GeV}^{-2}. \tag{7.87}$$

The slides emphasize a small but significant discrepancy with the beta-decay value $G\Big|_{\beta\text{-decay}} = 1.136(3) \times 10^{-5}\ \mathrm{GeV}^{-2}$, which motivates the next section.

7.12 Strange quarks, kaon decays, and Cabibbo mixing

7.12.1 Failure of universal $d \to u$ coupling for $s \to u$

Consider $K^+ \to e^+\nu_e$. The slides quote

$$\Gamma(K^+ \to e^+\nu_e) = 1.28 \times 10^3\ \mathrm{s}^{-1}. \tag{7.88}$$

If the s quark coupled to u with the same strength as d does, then (neglecting m_e)

$$\frac{\Gamma(K^+ \to e^+\nu_e)}{\Gamma(\pi^+ \to e^+\nu_e)} \simeq \frac{m_K f_K^2}{m_\pi f_\pi^2}. \tag{7.89}$$

Approximate chiral $SU_L(3) \times SU_R(3)$ suggests $f_K \simeq f_\pi$, leading to the rough expectation

$$\frac{\Gamma(K^+ \to e^+\nu_e)}{\Gamma(\pi^+ \to e^+\nu_e)}\bigg|_{\mathrm{th}} \simeq \frac{m_K}{m_\pi} \simeq 3.5. \tag{7.90}$$

But experimentally the ratio is $\simeq 0.27$. Conclusion:

$$\boxed{\text{The } s \text{ quark does not couple to } u \text{ with the same strength as } d \text{ does.}} \tag{7.91}$$

7.12.2 Introducing V_{ud} and V_{us} (Cabibbo idea)

Modify the current by introducing parameters V_{ud} and V_{us}:

$$J^\mu = \bar{\nu}_e \gamma^\mu P_L e + \bar{\nu}_\mu \gamma^\mu P_L \mu + V_{ud}\, \bar{u}\gamma^\mu P_L d + V_{us}\, \bar{u}\gamma^\mu P_L s + \cdots \tag{7.92}$$

Assuming $V_{ud} \simeq 1$, the kaon/pion ratio suggests $|V_{us}|^2 \sim 0.27$. A more accurate determination comes from $K^0 \to \pi^- e^+ \bar{\nu}_e$ (not depending on f_π or f_K):

$$|V_{us}| \simeq 0.2249(10). \tag{7.93}$$

Cabibbo parameterized

$$\sin\theta_c = V_{us}, \qquad \cos\theta_c = V_{ud}, \tag{7.94}$$

motivated by assuming that the quark fields appearing in the weak current are unitary rotations of the mass eigenstate fields:

$$\begin{pmatrix} d' \\ s' \end{pmatrix} = \begin{pmatrix} \cos\theta_c & \sin\theta_c \\ -\sin\theta_c & \cos\theta_c \end{pmatrix} \begin{pmatrix} d \\ s \end{pmatrix}, \qquad J^\mu = \bar{\nu}_e \gamma^\mu P_L e + \bar{\nu}_\mu \gamma^\mu P_L \mu + \bar{u}\gamma^\mu P_L d' + \cdots \tag{7.95}$$

7.12.3 More quarks and resolving the G discrepancy

A natural question raised in the slides: if only d' couples to u, who couples to s'? This motivates introducing a new quark c (charm) that couples to s':

$$J^\mu = \bar{\nu}_e \gamma^\mu P_L e + \bar{\nu}_\mu \gamma^\mu P_L \mu + \bar{u}\gamma^\mu P_L d' + \bar{c}\gamma^\mu P_L s' + \cdots \tag{7.96}$$

With Cabibbo's modification, the beta-decay determination is consistent with muon decay:

$$G_\beta \;=\; 1.136(3)\times 10^{-5}\,\mathrm{GeV}^{-2} \quad \| \quad G_\mu \cos\theta_c = 1.16632(2)\times 10^{-5}\,\mathrm{GeV}^{-2}\times 0.9744 = 1.13644\times 10^{-5}\,\mathrm{GeV}^{-2}. \tag{7.97}$$

So the discrepancy is resolved by the Cabibbo factor.

7.13 Counting parameters in quark mixing matrices: towards CKM

For N down-type quarks, the most general mixing is an $N \times N$ unitary matrix V with

$$\sum_i V_{ij} V_{ik}^* = \delta_{jk}. \tag{7.98}$$

The slides count the independent parameters:

- An arbitrary complex $N \times N$ matrix has N^2 moduli and N^2 phases.
- Unitarity imposes constraints:
  - N constraints for $j = k$ on moduli,
  - $N(N-1)/2$ constraints for $j \neq k$ on both moduli and phases.
- After imposing unitarity, the remaining are

$$\text{Moduli: } \frac{N(N-1)}{2}, \qquad \text{Phases: } \frac{N(N+1)}{2}. \tag{7.99}$$

- Additionally, $2N - 1$ phases can be absorbed by redefining quark fields.

Final count:

$$\boxed{\text{Independent moduli: } \frac{N(N-1)}{2}, \qquad \text{Independent phases: } \frac{(N-2)(N-1)}{2}.} \tag{7.100}$$

Checks:

- $N = 2$: one modulus, no phase (Cabibbo angle).
- $N = 3$: three moduli and **one phase** $\Rightarrow$ CP violation (Kobayashi–Maskawa).

For $N = 3$ the matrix is the **CKM matrix**.

7.14 CP violation from a complex CKM phase

For $N = 3$ families, a phase in the CKM matrix implies complex terms in the Fermi Lagrangian. Example term (if V_{ub} is complex):

$$V_{ub}V_{cs}^* \, \bar{u}\gamma^\mu P_L b \, \bar{s}\gamma_\mu P_L c + V_{ub}^* V_{cs} \, \bar{b}\gamma^\mu P_L u \, \bar{c}\gamma_\mu P_L s. \tag{7.101}$$

Under CP, one finds that the left current transforms into the right current (and vice versa), but the Lagrangian is invariant only if

$$V_{ub}V_{cs}^* = V_{ub}^* V_{cs}, \tag{7.102}$$

which fails if V_{ub} is complex. Hence CP is violated.

The slides highlight the historical narrative:

- CP violation was observed in neutral kaon decays since the 1960s.
- The idea that a third family could explain CP violation was realized before the b quark discovery.
- The discovery of b (1977) opened the possibility of explaining kaon CP violation in the electroweak framework.

7.15 Neutral kaons and CP violation (classic system)

Neutral kaons:

$$K^0 = (\bar{s}d), \qquad \bar{K}^0 = (d\bar{s}), \qquad J^P = 0^-. \tag{7.103}$$

Define CP eigenstates (using $C|\bar{K}^0\rangle = |K^0\rangle$ and $C|K^0\rangle = |\bar{K}^0\rangle$):

$$|K_S\rangle \equiv \frac{1}{\sqrt{2}}\left(|K^0\rangle - |\bar{K}^0\rangle\right), \qquad |K_L\rangle \equiv \frac{1}{\sqrt{2}}\left(|K^0\rangle + |\bar{K}^0\rangle\right), \tag{7.104}$$

with

$$CP|K_S\rangle = |K_S\rangle, \qquad CP|K_L\rangle = -|K_L\rangle. \tag{7.105}$$

Possible final states include $\pi\pi$ and $\pi\pi\pi$:

- ◇ Two-pion decay: the orbital angular momentum must be $L = 0$, and $CP|\pi\pi\rangle = +|\pi\pi\rangle$.
- ◇ Three-pion decay dominated by $L = 0$ gives $CP|\pi\pi\pi\rangle = -|\pi\pi\pi\rangle$.

Therefore, if CP were conserved:

$$|K_S\rangle \to |\pi\pi\rangle, \qquad |K_L\rangle \to |\pi\pi\pi\rangle. \tag{7.106}$$

Lifetimes:

$$\tau(K_S) \sim 8.9 \times 10^{-11}\ \text{s}, \qquad \tau(K_L) \sim 5.1 \times 10^{-8}\ \text{s}. \tag{7.107}$$

In a long-lived beam, essentially only K_L survive. Experimentally it was observed that K_L also decays to $\pi\pi$ with branching fractions

$$\text{BF}(K_L \to \pi^+\pi^-) \sim 2 \times 10^{-3}, \qquad \text{BF}(K_L \to \pi^0\pi^0) \sim 8 \times 10^{-4}, \tag{7.108}$$

which implies

$$\boxed{\text{CP is violated.}} \tag{7.109}$$

Consequences include particle–antiparticle asymmetries such as

$$\Gamma(K_L \to \pi^+ e^- \bar{\nu}_e) \neq \Gamma(K_L \to \pi^- e^+ \nu_e). \tag{7.110}$$

The slides also stress the cosmological relevance: CP violation is one of the three Sakharov conditions for explaining the matter–antimatter asymmetry.

7.16 T violation and CPT

The slides state:

- ◇ The same terms that violate CP in the Fermi Lagrangian also violate T (time reversal).
- ◇ Direct T violation has been observed (e.g. 1998 in K^0 decays, CPLEAR; 2012 in B decays, BaBar).
- ◇ CPT is always respected.

7.17 CKM matrix hierarchy and size of CP violation

The CKM matrix is known rather accurately. Its parametric structure is summarized as

$$V_{ij} = \begin{pmatrix} \mathcal{O}(1) & \mathcal{O}(\lambda) & \mathcal{O}(\lambda^3) \\ \mathcal{O}(\lambda) & \mathcal{O}(1) & \mathcal{O}(\lambda^2) \\ \mathcal{O}(\lambda^3) & \mathcal{O}(\lambda^2) & \mathcal{O}(1) \end{pmatrix}, \qquad \lambda \simeq 0.22, \tag{7.111}$$

with $j = (d, s, b)$ and $i = (u, c, t)$. Interpretation:

- ◇ If kinematically allowed, quarks prefer to decay within the same family.
- ◇ Next preference is to the closest family (controlled by powers of λ).

The size of CP violation can be estimated by Jarlskog's determinant,

$$\boxed{\text{Jarlskog invariant } \sim 2.96 \times 10^{-5}}, \tag{7.112}$$

while $|\det V| \sim 1$.

7.18 Neutrino masses, mixing, and oscillations (PMNS matrix)

The slides emphasize that the Standard Model has massless neutrinos, but oscillations observed in nature imply small neutrino masses.

Introducing a right-handed neutrino (which is sterile in the weak current) allows a Dirac mass term. Then, similarly to quarks, the neutrino fields appearing in the weak currents may not coincide with mass eigenstates, but be related by a unitary transformation. This leads to the leptonic analogue of CKM: the **Pontecorvo–Maki–Nakagawa–Sakata** (PMNS) matrix and to neutrino oscillations.

Observed oscillation evidence (as listed in the slides):

- ◇ Solar neutrinos (deficit observed since late 1960s; confirmed definitively by SNO in 2001).
- ◇ Atmospheric neutrinos (imbalance confirmed by Kamiokande in 1998).
- ◇ Reactor neutrinos.
- ◇ Beam neutrinos (accelerator-produced).

The PMNS matrix is summarized in the slides approximately as

$$U_{\text{PMNS}} \approx \begin{pmatrix} \sqrt{\frac{2}{3}} & \sqrt{\frac{1}{3}} & \mathcal{O}(\lambda) \\ -\sqrt{\frac{1}{6}} & \sqrt{\frac{1}{3}} & \sqrt{\frac{1}{2}} \\ \sqrt{\frac{1}{6}} & -\sqrt{\frac{1}{3}} & \sqrt{\frac{1}{2}} \end{pmatrix}, \qquad \lambda \simeq 0.22, \tag{7.113}$$

with a pattern very different from the quark sector, providing an additional possible source of CP violation.

If neutrino masses are Majorana:

- ◇ Two more phases appear ⇒ more sources of CP violation.
- ◇ One does not need a right-handed neutrino.
- ◇ Lepton number violation should appear, e.g. neutrinoless double beta decay (example in the slides: ${}^{128}_{52}\mathrm{Te} \to {}^{128}_{54}\mathrm{Xe}\, e^- e^-$).

7.19 Neutral currents: necessity, suppression of FCNC, and discovery

7.19.1 From charged currents to neutral currents

So far we have used a charged current J^μ that changes electric charge by one unit ($\Delta Q = +1$), and its conjugate $J^{\mu\dagger}$ with $\Delta Q = -1$:

$$J^\mu_{\rm cc} = \sum_F \bar{F}\,\tau^+\gamma^\mu P_L F', \qquad J^{\mu\dagger}_{\rm cc} = \sum_F \bar{F}'\,\tau^-\gamma^\mu P_L F, \tag{7.114}$$

where F runs over $SU(2)$ doublets such as

$$\begin{pmatrix}\nu_e\\ e\end{pmatrix}, \begin{pmatrix}u\\ d\end{pmatrix}, \begin{pmatrix}\nu_\mu\\ \mu\end{pmatrix}, \begin{pmatrix}c\\ s\end{pmatrix}, \ldots \tag{7.115}$$

and $F' = F$ for leptons, while for quarks $F' = VF$ with V the CKM matrix.

If an underlying $SU(2)$ structure exists, then besides $\tau^\pm$ one must also have τ^3, which leads to **neutral currents**.

7.19.2 Flavor-changing neutral currents (FCNC) are suppressed

Experimentally, neutral currents that change flavor are very suppressed, e.g.

$$\frac{\Gamma(K^0_L \to \mu^+\mu^-)}{\Gamma(K^0_L \to \text{all})} \simeq 9 \times 10^{-9}. \tag{7.116}$$

Therefore we only include neutral currents that do *not* change flavor. This is natural if CKM is unitary:

$$J^\mu_{\rm nc} \sim \sum_{F'} \bar{F}'\tau^3\gamma^\mu P_L F' = \sum_F \bar{F}\tau^3\gamma^\mu P_L F. \tag{7.117}$$

(While the full electroweak structure is more complicated, the key feature is invariance under unitary field redefinitions.)

7.19.3 Experimental discovery: Gargamelle (CERN, 1973)

Neutral currents were suggested (1958, Bludman) and incorporated in early electroweak ideas (Glashow, 1961). They do not change flavor, hence they compete with electromag-

netic interactions and are difficult to observe.

They were first detected at CERN in 1973 by the Gargamelle bubble chamber experiment. A (anti)neutrino beam was produced from decays $\pi^\pm, K^\pm \to \mu^\pm\nu$, with parent mesons produced by proton collisions on a Be target. Observed processes:

$$\bar{\nu}_\mu e^- \to \bar{\nu}_\mu e^-, \tag{7.118}$$
$$\bar{\nu}_\mu N \to \bar{\nu}_\mu X, \qquad (X \text{ hadrons, no muon}), \tag{7.119}$$
$$\nu_\mu N \to \nu_\mu X, \qquad (X \text{ hadrons, no muon}), \tag{7.120}$$

which are second order in the Fermi theory and hence suppressed. Nevertheless, the experiment found

$$R_\nu = \frac{\sigma_{\rm nc}(\nu_\mu N \to \nu_\mu X)}{\sigma_{\rm cc}(\nu_\mu N \to \mu^- X)} = 0.31 \pm 0.01, \qquad R_{\bar{\nu}} = \frac{\sigma_{\rm nc}(\bar{\nu}_\mu N \to \bar{\nu}_\mu X)}{\sigma_{\rm cc}(\bar{\nu}_\mu N \to \mu^+ X)} = 0.38 \pm 0.02, \tag{7.121}$$

showing that charged and neutral currents have roughly the same size.

7.20 Parametrizing neutral currents and extracting couplings from DIS

Neutral currents are parameterized in the slides as

$$J^\mu_{\rm nc} = \sum_f \bar{f}\,\gamma^\mu\,\frac{c_V^f - c_A^f\gamma^5}{2}\,f, \tag{7.122}$$

and the corresponding interaction is

$$\boxed{\mathcal{L}_{\rm nc} = 4\sqrt{2}\,G\,\rho\,J^{\mu\dagger}_{\rm nc}J_{{\rm nc}\,\mu}.} \tag{7.123}$$

Here f runs over fermion fields (quarks and leptons). The parameters ρ, c_V^f, c_A^f can be measured in high-energy neutrino deep inelastic scattering (DIS).

7.20.1 DIS kinematics and the parton model benchmark

Processes:

$$\nu_\ell N \to \nu_\ell X, \qquad \nu_\ell N \to \ell^- X, \tag{7.124}$$
$$\bar{\nu}_\ell N \to \bar{\nu}_\ell X, \qquad \bar{\nu}_\ell N \to \ell^+ X, \qquad \ell = e, \mu. \tag{7.125}$$

For electron DIS, the parton model gives (for $s \gg m_N$)

$$\frac{d\sigma}{dx\,dy} \simeq F_2(x)\,\frac{2\pi\alpha^2}{s}\,\frac{1}{q^4}\left(1 + (1-y)^2\right), \qquad F_2(x) = \sum_i f_i(x) Q_i^2\,x. \tag{7.126}$$

Kinematic variables as defined in the slides:

$$x = -\frac{q^2}{2m_N\nu}, \qquad \nu = \frac{p_N \cdot q}{m_N}, \qquad y = \frac{p_N \cdot q}{p_N \cdot p_A}, \qquad q = p_A - p_1, \tag{7.127}$$

where p_A is the incoming electron momentum and p_1 the outgoing electron momentum. From data, one extracts $F_2(x)$ and thus combinations of parton distributions (e.g. $xf_{uv}(x)$, $xf_{dv}(x)$, $xf_s(x)$).

7.20.2 Charged-current neutrino DIS formulas

Analogous formulas in the slides for charged-current neutrino DIS:

$$\frac{d\sigma(\nu_\ell N \to \ell^- X)}{dx\,dy} = \frac{G^2 s}{\pi}\left(xf_d(x) + xf_{\bar{u}}(x)(1-y)^2\right), \tag{7.128}$$

$$\frac{d\sigma(\bar{\nu}_\ell N \to \ell^+ X)}{dx\,dy} = \frac{G^2 s}{\pi}\left(xf_{\bar{d}}(x) + xf_u(x)(1-y)^2\right). \tag{7.129}$$

From these one can extract $xf_{uv}(x)$, $xf_{dv}(x)$ and $xf_s(x)$, i.e. how nucleons are made of partons.

7.20.3 Neutral-current DIS and extracted couplings

Similar formulas for $\nu_\ell N \to \nu_\ell X$ and $\bar{\nu}_\ell N \to \bar{\nu}_\ell X$ allow extraction of ρ, c_V^f, c_A^f and the parton charges Q_f. The slides report:

$$\rho \simeq 1, \tag{7.130}$$

and the following values:

f	c_V^f	c_A^f	Q_f
ν_ℓ	$1/2$	$1/2$	0
e^-	-0.03	$-1/2$	-1
u	0.19	$1/2$	$2/3$
d	-0.34	$-1/2$	$-1/3$

A remarkable relation holds:

$$\boxed{c_V^f = c_A^f - 2Q_f\,x, \qquad x \simeq 0.23.} \tag{7.131}$$

Therefore the neutral current can be written (as in the slides) as

$$\boxed{J^\mu_{\rm nc} = \sum_F \bar{F}\left(\frac{\tau^3}{2}\gamma^\mu P_L - x\,\bar{F}Q_F\gamma^\mu F\right),} \tag{7.132}$$

where Q_F is the charge matrix in each doublet:

$$Q_F = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix} \quad \text{for leptons,} \qquad Q_F = \begin{pmatrix} \frac{2}{3} & 0 \\ 0 & -\frac{1}{3} \end{pmatrix} \quad \text{for quarks.} \tag{7.133}$$

7.21 Big picture summary (within the slides' scope)

- ◇ Weak interactions are required by the extreme smallness of certain decay widths (pion, neutron, muon, strange hadrons) compared to typical strong widths.
- ◇ The Wu experiment established **parity violation**, motivating the **V–A** structure.
- ◇ The Fermi constant has dimension -2, indicating an effective, low-energy description consistent with massive vector exchange ($G \sim 1/m_W^2$).
- ◇ Low-energy processes:
  - $\pi^+ \to \ell^+ \nu_\ell$ exhibits **helicity suppression** ($\mathcal{M} \propto m_\ell$), yielding a parameter-free e/μ ratio.
  - Chiral symmetry and the non-linear sigma model allow a systematic hadronic representation of weak currents, giving $F_\pi = f_\pi$ and predicting other low-energy weak processes without new parameters at leading order.
  - Nucleon beta decay requires an axial coupling $g_A \simeq 1.27$ and leads, in the non-relativistic limit, to Fermi and Gamow–Teller operators.
- ◇ Muon decay provides an independent determination of G and motivates Cabibbo mixing to reconcile with beta decay determinations.
- ◇ Strange decays require different coupling strengths (V_{us} vs. V_{ud}), leading to Cabibbo mixing and, with three families, to the CKM phase and CP violation.
- ◇ Neutral currents are required by an underlying $SU(2)$ structure and were discovered in DIS; their couplings satisfy the relation $c_V^f = c_A^f - 2Q_f x$ with $x \simeq 0.23$.
- ◇ Neutrino masses and mixing lead to the PMNS matrix and neutrino oscillations, with possible additional CP-violating phases in the Majorana case.

8 . Electroweak Unification

8.1 Overview and strategy

The goal of electroweak unification is to build a *single* local gauge theory that:

- reproduces the observed **charged-current** weak interactions (mediated by $W^{\pm}$),
- reproduces **electromagnetism** (mediated by the photon A_μ),
- reproduces the observed **neutral-current** weak interactions (mediated by Z_μ),
- and explains why weak interactions are **short range** (massive vector bosons) while electromagnetism is **long range** (massless photon),
- all while keeping the **local gauge symmetry** exact in the Lagrangian.

We will proceed in three big steps:

1. Identify the observed currents (J^μ_{cc}, J^μ_{em}, J^μ_{nc}) and rewrite them in a form that suggests a gauge structure.
2. Build a Yang–Mills theory invariant under $SU_L(2) \times U_Y(1)$ and show how A_μ, $W^\pm_\mu$, Z_μ emerge after mixing.
3. Implement **spontaneous symmetry breaking** via the Higgs mechanism:

$$SU_L(2) \times U_Y(1) \longrightarrow U_{\text{em}}(1),$$

generating masses for $W^\pm$ and Z while keeping the photon massless.

8.2 Weinberg–Salam model for a single family

8.2.1 Fermion content and observed currents

We start with one family of leptons and quarks organized into left-handed doublets:

$$L = \begin{pmatrix} \nu_e \\ e \end{pmatrix}, \qquad Q = \begin{pmatrix} u \\ d \end{pmatrix}. \tag{8.1}$$

The experimentally observed weak charged current (V–A structure) reads

$$\boxed{J^\mu_{\rm cc} = \bar\nu_e \gamma^\mu P_L e + \bar u \gamma^\mu P_L d = \sum_{F=L,Q} \bar F\, \tau^+\, \gamma^\mu P_L F,} \tag{8.2}$$

where $P_L = \frac{1}{2}(1-\gamma_5)$ and τ^+ is the isospin raising operator acting in the doublet space. The electromagnetic current is vector-like (same coupling to left and right components):

$$\boxed{J^\mu_{\rm em} = \sum_f Q_f\, \bar f \gamma^\mu f,} \tag{8.3}$$

where the sum runs over the fermion fields $f = \nu_e, e, u, d$ (and later also right-handed singlets), and Q_f is the electric charge (in units of e).

Finally, the weak neutral current can be written in the general form

$$J^\mu_{\rm nc} = \sum_f \bar f \gamma^\mu \frac{c_V^f - c_A^f \gamma_5}{2} f \equiv \sum_f c_L^f\, \bar f \gamma^\mu P_L f + \sum_f c_R^f\, \bar f \gamma^\mu P_R f. \tag{8.4}$$

The coefficients satisfy

$$c_V^f = c_A^f - 2Q_f\, x, \qquad x \simeq 0.23, \qquad c_A^\nu = c_A^u = \frac{1}{2}, \quad c_A^e = c_A^d = -\frac{1}{2}, \tag{8.5}$$

and therefore

$$\boxed{c_L^f = c_A^f - Q_f x, \qquad c_R^f = -Q_f x.} \tag{8.6}$$

An immediate consequence is:

$$Q_f = 0 \implies c_R^f = 0 \quad \Rightarrow \quad \text{no right-handed neutral-current coupling.} \tag{8.7}$$

In particular, a *right-handed neutrino* (if introduced) would not interact through these currents.

8.2.2 A key identity: rewriting the neutral current

A crucial observation is that the weak neutral current can be expressed as

$$\boxed{J^\mu_{\rm nc} = \sum_{F=L,Q} \bar F\, \frac{\tau_3}{2} \gamma^\mu P_L F - x\, J^\mu_{\rm em}.} \tag{8.8}$$

This form is extremely suggestive: it separates a purely left-handed isospin piece (built from $\tau_3/2$) and a piece proportional to the electromagnetic current. This is precisely the structure that will appear after mixing two gauge fields into A_μ and Z_μ.

8.2.3 Hypercharge and the electromagnetic current

Let us focus on the *left-handed* part of the electromagnetic current. One can show

$$\sum_f Q_f\, \bar{f}\gamma^\mu P_L f = \sum_F \left(\bar{F}\, \frac{\tau_3}{2}\gamma^\mu P_L F + \bar{F}\, \frac{Y_F}{2}\gamma^\mu P_L F \right), \tag{8.9}$$

where F runs over the left-handed doublets L and Q, and Y_F is a new quantum number called **weak hypercharge**. For the two left-handed doublets in one family,

$$\boxed{\frac{Y_L}{2} = -\frac{1}{2}, \qquad \frac{Y_Q}{2} = \frac{1}{6}.} \tag{8.10}$$

The key point is that the hypercharge term is proportional to the identity in isospin space, so it commutes with τ_3 and with the charged current structure.

Including right-handed singlets, the full electromagnetic current becomes

$$J^\mu_{\text{em}} = \sum_F \bar{F}\, \frac{\tau_3}{2}\gamma^\mu P_L F + \sum_F \bar{F}\, \frac{Y_F}{2}\gamma^\mu P_L F + \sum_f Q_f\, \bar{f}\gamma^\mu P_R f. \tag{8.11}$$

From this, the neutral current can be reorganized into a form that separates:

- ◇ an isospin term,
- ◇ and a bracket containing the hypercharge (left-handed) and the right-handed charge terms.

Explicitly,

$$J^\mu_{\text{nc}} = (1-x)\sum_F \bar{F}\, \frac{\tau_3}{2}\gamma^\mu P_L F - x\left(\sum_F \bar{F}\, \frac{Y_F}{2}\gamma^\mu P_L F + \sum_f Q_f\, \bar{f}\gamma^\mu P_R f \right). \tag{8.12}$$

8.3 Quantum numbers: T_3, Q and weak hypercharge

8.3.1 Definition and table

We choose $SU_L(2)$ weak isospin with generator

$$T_3 = \frac{\tau_3}{2}.$$

Then the defining relation among electric charge, isospin and hypercharge is

$$\boxed{Q = T_3 + \frac{Y}{2}, \qquad \text{equivalently} \qquad \frac{Y}{2} = Q - T_3.} \tag{8.13}$$

For one family, the quantum numbers are:

Field	$SU_L(2)$	T_3	Q_f	$\frac{Y}{2} \equiv Q_f - T_3$
ν_{eL}	$\frac{1}{2}$	$+\frac{1}{2}$	0	$-\frac{1}{2}$
e_L	$\frac{1}{2}$	$-\frac{1}{2}$	-1	$-\frac{1}{2}$
u_L	$\frac{1}{2}$	$+\frac{1}{2}$	$\frac{2}{3}$	$\frac{1}{6}$
d_L	$\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{3}$	$\frac{1}{6}$
u_R	0	0	$\frac{2}{3}$	$\frac{2}{3}$
d_R	0	0	$-\frac{1}{3}$	$-\frac{1}{3}$
e_R	0	0	-1	-1

These assignments will be exactly the ones required for a consistent gauge theory under $SU_L(2) \times U_Y(1)$.

8.4 Building the electroweak gauge theory: $SU_L(2) \times U_Y(1)$

8.4.1 Yang–Mills Lagrangian and covariant derivatives

We now construct a local gauge theory invariant under

$$SU_L(2) \times U_Y(1).$$

Introduce gauge fields:

- $W_\mu = W_\mu^a T^a$ for $SU_L(2)$, with $T^a = \tau^a/2$,
- B_μ for $U_Y(1)$.

The Lagrangian for fermion kinetic terms plus gauge-field kinetic terms is

$$\boxed{\mathcal{L} = \sum_{F=L_L,Q_L} \bar{F}\, i\not{D}_F F + \sum_{f=u_R,d_R,e_R} \bar{f}\, i\not{D}_f f - \frac{1}{2}\mathrm{tr}(W_{\mu\nu}W^{\mu\nu}) - \frac{1}{4}B_{\mu\nu}B^{\mu\nu}.} \tag{8.14}$$

The covariant derivatives are:

$$\boxed{D_\mu^F = \partial_\mu + igW_\mu + ig'\frac{Y_F}{2}B_\mu, \qquad D_\mu^f = \partial_\mu + ig'\frac{Y_f}{2}B_\mu,} \tag{8.15}$$

where g and g' are the electroweak couplings.

The field strengths are:

$$W_{\mu\nu} = T^a W_{\mu\nu}^a, \qquad W_{\mu\nu}^a = \partial_\mu W_\nu^a - \partial_\nu W_\mu^a - g\,\epsilon^{abc}W_\mu^b W_\nu^c, \tag{8.16}$$

$$B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu, \tag{8.17}$$

with $[T^a, T^b] = i\epsilon^{abc}T^c$.

8.4.2 Gauge transformations

Under $SU_L(2) \times U_Y(1)$:

$$F(x) \to e^{ig'\frac{Y_F}{2}\theta(x)}\, g_L(x)\, F(x), \tag{8.18}$$

$$f(x) \to e^{ig'\frac{Y_f}{2}\theta(x)}\, f(x), \tag{8.19}$$

and the gauge fields transform as

$$W_\mu(x) \to g_L(x) W_\mu(x) g_L^\dagger(x) - \frac{i}{g}\, g_L(x)\partial_\mu g_L^\dagger(x), \tag{8.20}$$

$$B_\mu(x) \to B_\mu(x) - \partial_\mu \theta(x), \tag{8.21}$$

which implies

$$W_{\mu\nu} \to g_L W_{\mu\nu} g_L^\dagger, \qquad B_{\mu\nu} \to B_{\mu\nu}.$$

Thus the Lagrangian (8.14) is gauge invariant.

8.5 Physical gauge fields and currents

8.5.1 Charged fields $W_\mu^\pm$ and neutral mixing

Define the charged combinations

$$\boxed{W_\mu^\pm = \frac{1}{\sqrt{2}}\left(W_\mu^1 \mp i W_\mu^2\right).} \tag{8.22}$$

The neutral fields B_μ and W_μ^3 mix into the physical photon A_μ and Z_μ through the Weinberg angle θ_W:

$$A_\mu = \cos\theta_W\, B_\mu + \sin\theta_W\, W_\mu^3, \tag{8.23}$$

$$Z_\mu = -\sin\theta_W\, B_\mu + \cos\theta_W\, W_\mu^3, \tag{8.24}$$

or inverted,

$$B_\mu = \cos\theta_W\, A_\mu - \sin\theta_W\, Z_\mu, \tag{8.25}$$

$$W_\mu^3 = \sin\theta_W\, A_\mu + \cos\theta_W\, Z_\mu, \tag{8.26}$$

with $s_W \equiv \sin\theta_W$, $c_W \equiv \cos\theta_W$.

8.5.2 Charged current interaction

From the covariant derivative, the charged-current interaction is

$$\mathcal{L}_{\rm cc} = -\sum_{F=L_L,Q_L} g\,\bar{F}\gamma^\mu(T^1W^1_\mu + T^2W^2_\mu)F \tag{8.27}$$

$$= -\sum_{F=L_L,Q_L} \frac{g}{\sqrt{2}}\bar{F}\gamma^\mu(\tau^+W^+_\mu + \tau^-W^-_\mu)F \tag{8.28}$$

$$= \boxed{-\frac{g}{\sqrt{2}}\left(J^\mu_{\rm cc}W^+_\mu + J^{\dagger\mu}_{\rm cc}W^-_\mu\right).} \tag{8.29}$$

8.5.3 Electromagnetism and the fundamental relation $gs_W = g'c_W = e$

The electromagnetic interaction comes from the A_μ piece inside W^3_μ and B_μ. Requiring that **left and right** components couple in the *same way* to A_μ enforces:

$$eQ_f = g'\frac{Y_f}{2}c_W = gT_3s_W + g'\frac{Y_F}{2}c_W. \tag{8.30}$$

Using $\frac{Y}{2} = Q - T_3$, one arrives at the central electroweak relation

$$\boxed{g\,s_W = g'\,c_W = e.} \tag{8.31}$$

This ensures that the photon couples universally to electric charge and that QED emerges as the unbroken gauge symmetry after symmetry breaking.

8.5.4 Neutral current interaction and g_Z

Inserting the Z_μ combinations yields the Z interaction in the compact form

$$\mathcal{L}_{\rm nc} = -g_Z\,Z_\mu\,J^\mu_{\rm nc}, \qquad \boxed{g_Z \equiv \frac{e}{c_W s_W}.} \tag{8.32}$$

The neutral current can be written as

$$\boxed{J^\mu_{\rm nc} = \sum_{F=L_L,Q_L}\bar{F}\gamma^\mu T_3F - s_W^2\sum_{f=u,d,e}Q_f\,\bar{f}\gamma^\mu f,} \tag{8.33}$$

which matches the previously identified structure (8.8) with $x = s_W^2$.

8.6 The Higgs mechanism

8.6.1 The problem: massless gauge bosons

The gauge-invariant Lagrangian constructed so far contains only *massless* vector bosons. This would imply long-range weak forces, contradicting observation: weak interactions are extremely short range.

Adding explicit mass terms "by hand" is *not allowed* because such terms would break the local gauge symmetry $SU_L(2) \times U_Y(1)$.

8.6.2 The solution: spontaneous symmetry breaking

We use spontaneous symmetry breaking: the Lagrangian is fully symmetric, but the vacuum is not. The mechanism requires a scalar multiplet with a non-zero vacuum expectation value (VEV).

We need at least one *neutral* component to acquire a VEV. Since

$$Q = T_3 + \frac{Y}{2},$$

a neutral component must satisfy $Q = 0$. Taking a scalar doublet with $Y/2 = +1/2$ achieves this:

$$\boxed{\phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix}, \qquad \frac{Y_\phi}{2} = \frac{1}{2}.} \tag{8.34}$$

Its gauge transformation is

$$\phi(x) \longrightarrow g_L(x)\, e^{ig'\theta(x)/2}\, \phi(x). \tag{8.35}$$

8.6.3 Vacuum structure and unitary gauge

Assume a potential $V(\phi^\dagger\phi)$ with a minimum at

$$\boxed{\phi^\dagger\phi = \frac{v^2}{2} \in \mathbb{R}_+.} \tag{8.36}$$

A general field configuration near the minimum can be parameterized as

$$\phi(x) = U(x)\, \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix}, \qquad U(x) \in SU(2). \tag{8.37}$$

Important points:

- ◇ The complex doublet ϕ has **4 real degrees of freedom**.
- ◇ $h(x)$ is a real field describing fluctuations *along* the radial direction.
- ◇ $U(x)$ contains the would-be Goldstone bosons if the symmetry were global.

- ◇ Since the symmetry is *local*, $U(x)$ can be removed by a gauge transformation: this is the **unitary gauge**.

In unitary gauge,

$$\boxed{\phi(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix}.} \tag{8.38}$$

8.6.4 Unbroken subgroup and breaking pattern

The vacuum configuration $h(x) = 0$ is invariant under transformations with

$$g_L(x) = e^{ig'\theta(x)T_3},$$

which corresponds precisely to the electromagnetic subgroup. Hence the breaking pattern is:

$$\boxed{SU_L(2) \times U_Y(1) \longrightarrow U_{\text{em}}(1).} \tag{8.39}$$

8.6.5 Gauge boson masses from the Higgs kinetic term

The scalar kinetic term is

$$(D_\mu\phi)^\dagger D^\mu\phi, \qquad D_\mu = \partial_\mu + igW_\mu + ig'\frac{1}{2}B_\mu. \tag{8.40}$$

In terms of physical fields it can be written (as given) as

$$\begin{aligned} D_\mu &= \partial_\mu + i\frac{g}{\sqrt{2}}(\tau^+W_\mu^+ + \tau^-W_\mu^-) + igT_3W_\mu^3 + ig'\frac{1}{2}B_\mu \\ &= \partial_\mu + i\frac{g}{\sqrt{2}}(\tau^+W_\mu^+ + \tau^-W_\mu^-) + \begin{pmatrix} ieA_\mu + ig_Z(c_W^2 - s_W^2)Z_\mu & 0 \\ 0 & -ig_Z\frac{1}{2}Z_\mu \end{pmatrix}. \end{aligned} \tag{8.41}$$

Now evaluate at the vacuum configuration ($h(x) = 0$):

$$\phi(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix}.$$

Then

$$D_\mu\phi = \begin{pmatrix} i\frac{gv}{2}W_\mu^+ \\ -i\frac{g_Zv}{2\sqrt{2}}Z_\mu \end{pmatrix}, \tag{8.42}$$

and therefore

$$(D_\mu\phi)^\dagger D^\mu\phi = \frac{g^2v^2}{4}W_\mu^-W^{+\mu} + \frac{g_Z^2v^2}{8}Z_\mu Z^\mu. \tag{8.43}$$

This is precisely a mass term for $W^\pm$ and Z. Reading off the masses,

$$\boxed{m_W = \frac{gv}{2}, \qquad m_Z = \frac{g_Zv}{2}, \qquad g = g_Zc_W \;\Rightarrow\; m_Z > m_W.} \tag{8.44}$$

The photon remains massless because $U_{\text{em}}(1)$ is unbroken.

8.6.6 Degrees of freedom reshuffling: "eating" Goldstones

Before symmetry breaking:

- ◇ 3 massless gauge bosons $\Rightarrow 3 \times 2 = 6$ d.o.f.,
- ◇ 2 complex scalars $\Rightarrow 2 \times 2 = 4$ d.o.f.,

total 10 degrees of freedom.

After symmetry breaking:

- ◇ 3 massive gauge bosons ($W^{\pm}$, Z) $\Rightarrow 3 \times 3 = 9$ d.o.f.,
- ◇ 1 real scalar (h) $\Rightarrow 1$ d.o.f.,

total again 10.

Thus, the would-be Goldstone bosons are removed by a gauge transformation and become the longitudinal polarizations of the massive vector bosons: this is the **Higgs mechanism**. The remaining scalar field $h(x)$ describes a physical neutral scalar particle: the **Higgs boson**.

8.6.7 Historical note (as stated)

Some physicists speculated the field $h(x)$ could be superfluous (no Higgs particle), while others generalized to several scalar multiplets (several Higgs bosons). The Higgs boson was found at the LHC (CERN) in 2013, and so far no other fundamental scalar particle has showed up.

8.7 Matching to Fermi theory and numerical predictions

At low energy $E \ll m_W, m_Z$, amplitudes must reproduce the effective four-fermion Fermi theory. Matching yields:

$$\boxed{\frac{g^2}{2m_W^2} = 2\sqrt{2}\,G, \qquad \frac{g_Z^2}{m_Z^2} = 4\sqrt{2}\,\rho\,G.} \tag{8.45}$$

From these equalities one obtains

$$\boxed{\rho = \frac{g_Z^2}{m_Z^2}\frac{m_W^2}{g^2} = 1, \qquad v = \frac{1}{\sqrt{\sqrt{2}G}} \simeq 246 \text{ GeV}.} \tag{8.46}$$

Using $s_W^2 \simeq 0.23$ from low-energy experiments, one gets

$$g = \frac{e}{s_W} \simeq 0.63, \qquad g' = \frac{e}{c_W} \simeq 0.34, \qquad g_Z = \frac{e}{s_W c_W} \simeq 0.72, \tag{8.47}$$

and the corresponding (tree-level) mass estimates:

$$m_W \simeq 78 \text{ GeV}, \qquad m_Z \simeq 89 \text{ GeV}. \tag{8.48}$$

This constituted a prediction for m_W and m_Z, which were found at CERN in 1983.

The current experimental values quoted are:

$$m_W \simeq 80.379(12) \text{ GeV}, \qquad m_Z \simeq 91.1876(21) \text{ GeV}. \tag{8.49}$$

8.8 Fermion masses from Yukawa terms

8.8.1 Why fermions are massless without Yukawas

So far, we have generated masses for the vector bosons while preserving gauge symmetry in the Lagrangian. However, fermions would remain massless unless we add additional gauge-invariant terms.

8.8.2 Gauge-invariant Yukawa interactions

The scalar doublet ϕ allows **Yukawa terms**:

$$\boxed{\mathcal{L}_{\text{Yuk}} = -\lambda_e \, \bar{L} \, \phi \, e_R - \lambda_d \, \bar{Q} \, \phi \, d_R - \lambda_u \, \bar{Q} \, \phi^c \, u_R + \text{H.c.}} \tag{8.50}$$

where

$$\boxed{\phi^c \equiv i\tau_2 \, \phi^*, \qquad \phi^c(x) \longrightarrow g_L(x) e^{-i\theta(x)/2} \, \phi(x).} \tag{8.51}$$

In unitary gauge,

$$\phi(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix}, \qquad \phi^c(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} v + h(x) \\ 0 \end{pmatrix}.$$

Substituting into (8.50) gives

$$\mathcal{L}_{\text{Yuk}} = -\left(\lambda_e \, \bar{e}_L e_R + \lambda_d \, \bar{d}_L d_R + \lambda_u \, \bar{u}_L u_R\right) \frac{1}{\sqrt{2}}(v + h) + \text{H.c.} \tag{8.52}$$

This splits into a mass term and a Higgs–fermion interaction:

$$\boxed{m_f = \frac{\lambda_f v}{\sqrt{2}}, \qquad f = e, d, u,} \tag{8.53}$$

and

$$\boxed{\mathcal{L}_{h\bar{f}f} = -\left(m_e \bar{e}_L e_R + m_d \bar{d}_L d_R + m_u \bar{u}_L u_R + \text{H.c.}\right) \frac{h}{v}.} \tag{8.54}$$

Thus, **the Higgs coupling to a fermion is proportional to the fermion mass**.

8.9 Higgs interactions with vector bosons and self-interactions

8.9.1 Vector boson couplings

From the Higgs kinetic term one obtains, as given,

$$(D_\mu\phi)^\dagger D^\mu\phi = \frac{1}{2}\partial_\mu h\,\partial^\mu h + m_W^2 W_\mu^- W^{+\mu}\left(1+\frac{h}{v}\right)^2 + \frac{m_Z^2}{2}Z_\mu Z^\mu\left(1+\frac{h}{v}\right)^2. \tag{8.55}$$

This explicitly shows:

- the Higgs kinetic term $\frac{1}{2}(\partial h)^2$,
- the W and Z mass terms (from the 1 inside $(1+h/v)^2$),
- and the Higgs couplings to W and Z (from the linear and quadratic terms in h/v).

Since $v \simeq 246$ GeV, these couplings are typically small, except for the top quark ($m_t \simeq 173$ GeV) and for the heavy gauge bosons $W^\pm$ and Z^0.

8.9.2 Higgs self-interaction from the potential

The potential $V(\phi^\dagger\phi)$ with a non-zero minimum produces a Higgs mass and self-interactions. Using the stated analogy with the linear sigma model, one may write:

$$\boxed{\mathcal{L}_h = -\frac{m_h^2}{2}h^2 - \frac{m_h^2}{2v}h^3 - \frac{m_h^2}{8v^2}h^4.} \tag{8.56}$$

Here m_h was initially a free parameter (hardly constrained by other observables). Once m_h is measured, the Higgs self-interactions are fixed.

The quoted current Higgs mass value is:

$$m_h = 125.10(14)\ \text{GeV}. \tag{8.57}$$

8.10 More families and flavor structure

8.10.1 Generalizing the fermion content

In nature there are at least three families. For an arbitrary number of families N_f:

$$L_i = \begin{pmatrix}\nu_{l_i}\\ l_i\end{pmatrix}, \qquad Q_i = \begin{pmatrix}u_i\\ d_i\end{pmatrix}, \qquad i = 1,\ldots,N_f, \tag{8.58}$$

with examples:

$$(L_1,Q_1) = \left(\begin{pmatrix}\nu_e\\ e\end{pmatrix},\begin{pmatrix}u\\ d\end{pmatrix}\right), \quad (L_2,Q_2) = \left(\begin{pmatrix}\nu_\mu\\ \mu\end{pmatrix},\begin{pmatrix}c\\ s\end{pmatrix}\right), \quad (L_3,Q_3) = \left(\begin{pmatrix}\nu_\tau\\ \tau\end{pmatrix},\begin{pmatrix}t\\ b\end{pmatrix}\right).$$

The gauge-field part of the Lagrangian is unchanged. The fermion kinetic terms become

$$\mathcal{L}_{\text{kin}}^{\text{fermions}} = \sum_{i=1}^{N_f} \left(\sum_{F_i = L_{iL}, Q_{iL}} \bar{F}_i \, i \not{D}_F F_i + \sum_{f_i = u_{iR}, d_{iR}, e_{iR}} \bar{f}_i \, i \not{D}_f f_i \right). \tag{8.59}$$

8.10.2 General Yukawa matrices and mass diagonalization

With several families, Yukawa terms become matrices in family space:

$$\boxed{\mathcal{L}_{\text{Yuk}} = \sum_{i,j=1}^{N_f} \left[-\lambda_{ij}^{l} \, \bar{L}_{iL} \phi \, l_{jR} - \lambda_{ij}^{d} \, \bar{Q}_{iL} \phi \, d_{jR} - \lambda_{ij}^{u} \, \bar{Q}_{iL} \phi^c \, u_{jR} + \text{H.c.} \right].} \tag{8.60}$$

This leads to **off-diagonal mass terms**: fields with well-defined $SU_L(2) \times U_Y(1)$ quantum numbers are not mass eigenstates.

To obtain mass eigenstates, one diagonalizes the mass matrices through **unitary** field redefinitions (to keep canonical kinetic terms). For down-type quarks, write the mass term schematically as

$$\bar{d}_L \, \Lambda_d \, d_R = \sum_{i,j} \bar{d}_{iL} \, \Lambda_{ij}^{d} \, d_{jR}, \qquad \boxed{\Lambda_{ij}^{d} \equiv \lambda_{ij}^{d} \frac{v}{\sqrt{2}}.} \tag{8.61}$$

Assuming $\det \Lambda_d \neq 0$, one writes Λ_d as a unitary times a Hermitian matrix:

$$\boxed{\Lambda_d = U_d M_d, \qquad M_d = S_d D_d^{1/2} S_d^\dagger, \qquad \Lambda_d^\dagger \Lambda_d = S_d D_d S_d^\dagger,} \tag{8.62}$$

where S_d is unitary and D_d is diagonal positive definite (from diagonalizing the Hermitian matrix $\Lambda_d^\dagger \Lambda_d$). One can verify the unitarity of U_d as stated:

$$U_d U_d^\dagger = \mathbf{1}.$$

Then the down-quark mass term becomes diagonal under

$$\boxed{d_R \to S_d \, d_R, \qquad d_L \to U_d S_d \, d_L,} \tag{8.63}$$

where d_R and d_L are vectors in family space (indices suppressed).

One proceeds analogously for up-type quarks and charged leptons.

8.10.3 Why neutral currents remain diagonal, and charged currents do not

Because the transformations are unitary, they do *not* affect terms that are diagonal in isospin space, namely:

- ◇ the electromagnetic current,
- ◇ the neutral current.

However, the charged current involves u_L and d_L together. Since the unitary transformations for up and down quarks need not be the same, the charged current acquires a mixing matrix:

$$\boxed{\bar{u}_L \gamma^\mu d_L \;\longrightarrow\; \bar{u}_L \gamma^\mu V_{\rm CKM}\, d_L, \qquad V_{\rm CKM} \equiv S_u^\dagger U_u^\dagger U_d S_d.} \tag{8.64}$$

This is the origin of the **CKM matrix** in the electroweak theory.

8.10.4 Leptons and the PMNS comment

For leptons, the charged current transforms as

$$\bar{\nu}_{lL} \gamma^\mu l_L \;\longrightarrow\; \bar{\nu}_{lL} \gamma^\mu U_l S_l\, l_L. \tag{8.65}$$

Since there is no neutrino mass term in this setup, one can perform a unitary transformation on the neutrino fields to diagonalize the charged current. If one includes a right-handed neutrino, then a neutrino mass term becomes possible and the situation becomes analogous to the quark case: a **PMNS matrix** arises.

8.11 The GIM mechanism and suppression of FCNC

8.11.1 The empirical fact

Flavor-changing neutral currents (FCNC) are very suppressed. For instance,

$$\frac{\Gamma(K_L^0 \to \mu^+\mu^-)}{\Gamma(K_L^0 \to \text{all})} \simeq 9 \times 10^{-9}. \tag{8.66}$$

A naive expectation from weak interactions might suggest a suppression of order

$$\alpha_W^2, \qquad \alpha_W \equiv \frac{g^2}{4\pi} \simeq 0.032, \qquad \alpha_W^2 \simeq 10^{-3},$$

which is *far* too weak compared with 10^{-9}.

8.11.2 Origin: unitarity of the CKM matrix (GIM)

The strong suppression arises from the unitarity of the CKM matrix: this is the **GIM mechanism** (Glashow–Iliopoulos–Maiani, 1970).

To illustrate, consider two families with Cabibbo mixing:

$$\begin{pmatrix} d' \\ s' \end{pmatrix} = \begin{pmatrix} \cos\theta_c & \sin\theta_c \\ -\sin\theta_c & \cos\theta_c \end{pmatrix} \begin{pmatrix} d \\ s \end{pmatrix}. \tag{8.67}$$

Then the charged current is

$$J^\mu = \bar{\nu}_e \gamma^\mu P_L e + \bar{\nu}_\mu \gamma^\mu P_L \mu + \bar{u} \gamma^\mu P_L d' + \bar{c} \gamma^\mu P_L s'. \tag{8.68}$$

For $K_L^0 \to \mu^+\mu^-$ there are two loop diagrams contributing (as shown in the slides). If one neglects $m_u, m_c \ll m_W$, the two contributions cancel.

Including quark masses, the cancellation is not exact and produces an *extra* suppression:

$$\boxed{\mathcal{A}_{\text{FCNC}} \sim \frac{m_c^2}{m_W^2} \sim 3.2 \times 10^{-4},} \tag{8.69}$$

and therefore for the decay width,

$$\boxed{\Gamma_{\text{FCNC}} \sim \left(\frac{m_c^2}{m_W^2}\right)^2 = \frac{m_c^4}{m_W^4} \sim 10^{-7}.} \tag{8.70}$$

This explains why FCNC processes can be far more suppressed than a naive α_W^2 estimate.

8.11.3 Third family contribution and CKM hierarchy

Including the third family adds a contribution involving the top quark:

- since $m_t > m_W$, it could be potentially large,
- but the relevant CKM elements are small:

$$|V_{ts}| \sim \lambda^2, \qquad |V_{td}| \sim \lambda^3, \qquad \lambda \sim 0.22,$$

so the contribution is suppressed by

$$\boxed{\sim \lambda^5 \sim 5 \times 10^{-4},} \tag{8.71}$$

a suppression comparable in size to the two-family charm mechanism. Similar (not identical) suppressions occur in $D^0 \to \mu^+\mu^-$, $B^0 \to \mu^+\mu^-$ and non-leptonic FCNC decays.

Since FCNC are very suppressed within the electroweak theory, they provide a good place to search for effects of physics beyond the Standard Model.

8.12 Electroweak phenomenology

8.12.1 $W^\pm$ leptonic decays and lepton universality

Consider

$$W_A^+ \to l_1^+ \, \nu_{l\,2}.$$

The relevant interaction is

$$\mathcal{L}_{\text{int}} = -\frac{g}{\sqrt{2}} J_{\text{cc}}^\mu W_\mu^+, \qquad J_{\text{cc}}^\mu \simeq \bar{\nu}_l \gamma^\mu P_L l. \tag{8.72}$$

The matrix element is

$$\mathcal{M} = -\frac{g}{\sqrt{2}} \bar{u}(2)\gamma^\mu P_L v(1)\, \epsilon_\mu(A). \tag{8.73}$$

Neglecting lepton masses and averaging over W^+ polarizations, the partial width is

$$\boxed{\Gamma(W^+ \to l^+\nu_l) = \frac{\alpha_W\, m_W}{12}.} \tag{8.74}$$

Numerically this matches the quoted values:

$$\Gamma_{\text{Exp}} \simeq 0.22 \text{ GeV}, \qquad \Gamma_{\text{Th}} \simeq 0.21 \text{ GeV}.$$

The result is the same for any light lepton family: **lepton universality**.

8.12.2 $W^\pm$ hadronic decays, CKM suppression, and colors

Using the duality hypothesis:

$$\Gamma(W^+ \to \text{hadrons}) = \Gamma(W^+ \to \text{quarks}).$$

The quark-level channels are

$$\begin{aligned}\Gamma(W^+ \to \text{quarks}) = \Gamma(W^+ \to u\bar{d}) + \Gamma(W^+ \to u\bar{s}) + \Gamma(W^+ \to u\bar{b}) \\ + \Gamma(W^+ \to c\bar{d}) + \Gamma(W^+ \to c\bar{s}) + \Gamma(W^+ \to c\bar{b}).\end{aligned} \tag{8.75}$$

The CKM hierarchy implies suppressions such as

$$\Gamma(W^+ \to u\bar{s}) \sim |V_{us}|^2 \sim \lambda^2,$$

$$\Gamma(W^+ \to u\bar{b}) \sim |V_{ub}|^2 \sim \lambda^6,$$
$$\Gamma(W^+ \to c\bar{d}) \sim |V_{cd}|^2 \sim \lambda^2,$$
$$\Gamma(W^+ \to c\bar{b}) \sim |V_{cb}|^2 \sim \lambda^4.$$

The notes emphasize that QCD corrections are of order $\alpha_s(m_Z) \sim 0.11$, larger than $\lambda^2 \sim 0.048$. Therefore, at leading order (neglecting quark masses),

$$\Gamma(W^+ \to \text{hadrons}) \simeq \Gamma(W^+ \to u\bar{d}) + \Gamma(W^+ \to c\bar{s}). \tag{8.76}$$

Each quark channel has an extra factor $N_c = 3$ (number of colors), so:

$$\boxed{\Gamma(W^+ \to \text{hadrons}) \simeq 2 \times N_c \times \Gamma(W^+ \to l^+\nu_l) = \frac{\alpha_W m_W N_c}{6}.} \tag{8.77}$$

Numerically:

$$\Gamma_{\text{Exp}} \simeq 1.4 \text{ GeV}, \qquad \Gamma_{\text{Th}} \simeq 1.3 \text{ GeV}.$$

8.12.3 Z^0 decays: couplings and partial widths

The interaction is

$$\mathcal{L}_{\text{int}} = -g_Z\, J^\mu_{\text{nc}} Z_\mu, \qquad g_Z = \frac{g}{c_W}. \tag{8.78}$$

For each fermion:

$$J^\mu_{\rm nc} \simeq C^f_L\, \bar f \gamma^\mu P_L f + C^f_R\, \bar f \gamma^\mu P_R f, \tag{8.79}$$

with

$$\boxed{C^f_L = T_3 - Q_f s^2_W, \qquad C^f_R = -Q_f s^2_W.} \tag{8.80}$$

Since $m_f \ll m_Z$ for all fermions except the top quark ($m_t > m_Z$), one neglects the fermion masses. Then left and right decays decouple and the partial widths are:

$$\boxed{\Gamma(Z^0 \to l_L \bar l_R) = \frac{\alpha_W m_Z}{6c^2_W}\,(C^l_L)^2, \qquad \Gamma(Z^0 \to l_R \bar l_L) = \frac{\alpha_W m_Z}{6c^2_W}\,(C^l_R)^2,} \tag{8.81}$$

$$\boxed{\Gamma(Z^0 \to q_L \bar q_R) = \frac{\alpha_W m_Z N_c}{6c^2_W}\,(C^q_L)^2, \qquad \Gamma(Z^0 \to q_R \bar q_L) = \frac{\alpha_W m_Z N_c}{6c^2_W}\,(C^q_R)^2.} \tag{8.82}$$

8.12.4 Total Z^0 width and the number of light neutrinos

Summing over all accessible fermions gives

$$\boxed{\begin{aligned}\Gamma(Z^0 \to \text{all}) = \frac{\alpha_W m_Z}{6c^2_W}\Bigg[& N_\nu \left(\frac{1}{2}\right)^2 \\ & + 3\left(\left(-\frac{1}{2} + s^2_W\right)^2 + s^4_W\right) \\ & + 2N_c\left(\left(\frac{1}{2} - \frac{2s^2_W}{3}\right)^2 + \left(-\frac{2s^2_W}{3}\right)^2\right) \\ & + 3N_c\left(\left(-\frac{1}{2} + \frac{s^2_W}{3}\right)^2 + \left(\frac{s^2_W}{3}\right)^2\right)\Bigg].\end{aligned}}$$

Here N_ν is the number of neutrinos with mass $m_\nu < m_Z/2$. Therefore an accurate measurement of $\Gamma(Z^0 \to \text{all})$ determines the number of *light* neutrinos, and hence the number of families in the electroweak theory.

Using $m_Z \simeq 91.2$ GeV, $\alpha_W \simeq 0.0336$ and $s^2_W \simeq 0.23$, the quoted values (in GeV) are:

N_ν	EW	EW+QCD	Exp
2	2.26	2.38	–
3	2.42	2.55	2.4952(23)
4	2.59	2.71	–

The LEP measurement (00) thus points to $N_\nu = 3$ light neutrinos.

8.13 Summary of boxed key results

For quick reference, the most important relations established in this chapter are:

- ◇ Charge–hypercharge relation:

$$\boxed{Q = T_3 + \frac{Y}{2}.}$$

- ◇ Electroweak coupling relation defining the electric charge:

$$\boxed{g\, s_W = g'\, c_W = e.}$$

- ◇ Neutral coupling:

$$\boxed{g_Z = \frac{e}{c_W s_W}.}$$

- ◇ Higgs breaking pattern and gauge boson masses:

$$\boxed{SU_L(2) \times U_Y(1) \to U_{\rm em}(1), \qquad m_W = \frac{gv}{2}, \quad m_Z = \frac{g_Z v}{2}.}$$

- ◇ Matching to Fermi theory (tree level):

$$\boxed{\frac{g^2}{2m_W^2} = 2\sqrt{2}\, G, \quad \rho = 1, \quad v \simeq 246 \text{ GeV}.}$$

- ◇ Fermion masses from Yukawas:

$$\boxed{m_f = \frac{\lambda_f v}{\sqrt{2}}, \qquad \mathcal{L}_{h\bar{f}f} \propto \frac{m_f}{v}\, h.}$$

- ◇ CKM matrix in the charged current:

$$\boxed{\bar{u}_L \gamma^\mu d_L \;\to\; \bar{u}_L \gamma^\mu V_{\rm CKM} d_L.}$$

- ◇ Representative electroweak widths:

$$\boxed{\Gamma(W \to l\nu) = \frac{\alpha_W m_W}{12}, \qquad \Gamma(W \to \text{hadrons}) \simeq \frac{\alpha_W m_W N_c}{6}.}$$

- ◇ Total Z width and N_ν:

$$\boxed{\Gamma(Z \to \text{all}) \text{ fixes } N_\nu \text{ (number of light neutrinos).}}$$

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